

© Reda Allah Hasan Zayed, 2025
Licensed under CC BY-NC-ND 4.0
github.com/Rezaa590/student-performance-project

Applied Regression Analysis - STAT 2240
Regression Analysis of Factors Influencing Student Academic Performances
Phase II Report

Dr Yahia Al-Tohamy and TA Nermine Mamdouh

Faculty of Computational Sciences and Mathematics

By: Jasmine Taha, Kosisochukwu Onyekwulunne, Reda Allah Hasan Zayed

December 4th, 2025

Table of Contents

Introduction	2
Methodology-Literature Review	4
Literature Review	4
Methodology	5
Descriptive Statistics.....	8
Qualitative Variables	8
Frequency and Mode Tables	16
Quantitative Variables	18
Summary Table of Quantitative Variables.....	24
Correlation Matrix of all Variables	28
Inferential Statistics	30
Step 1: Multicollinearity Diagnostics	30
Step 2: Linearity	31
Step 3: Fitting the Multiple Regression Model.....	32
Checking for extreme values	33
Checking the Model Assumptions	35
• Endogeneity	35
Checking assumptions related to the residuals.....	37
• Normality of Errors	38
Checking for the Presence of Autocorrelation.....	39
Model Selection.....	40
Methods of model selection.....	40
1. Backward Elimination	40
2. Forward Selection.....	41
3. Stepwise Selection	42
Identify Best Model.....	42
• Comparing.....	42
• The best-fitted model.....	43
Final Model Reporting	44
Coefficient Interpretation	45
Statistical Analysis of the Model	47
Hypothesis Tests.....	47
ANOVA TABLE	53
Confidence Intervals.....	53
Conclusion	54

Introduction

Overview of the topic /purpose of studying the dataset:

This dataset examines students' final exam scores in relation to 20 personal, social, and educational factors, with 6,607 observations and 20 attributes tested. The final exam score serves as the primary measure of academic performance. The purpose of this report is to analyze how these factors influence student achievement and to develop a regression model capable of predicting exam performance. Specifically, the analysis aims to determine which variables have the strongest impact on final exam results, how personal and educational behaviors contribute to academic outcomes, and whether the dataset can be used to build a reliable predictive model. Through a combination of descriptive analysis and regression techniques, this project seeks to provide a deeper understanding of the determinants of academic success.

The description of the dataset predictor variables:

This dataset consists of 6,602 student records and includes 20 predictor variables capturing aspects of studying habits, family context, school conditions, lifestyle patterns, and others. These variables are used to examine their influence on students' academic achievement, represented by the target variable, final exam score. The dataset was obtained from the Kaggle platform <https://www.kaggle.com/ayeshaseherr/student-performance>.

There are 7 quantitative predictor variables(continuous, discrete) in the examined dataset, which are:

- **Hours_studied:** continuous, Number of hours the student reported studying per week.
- **Attendance:** continuous, Percentage of classes attended.
- **Tutoring_Sessions:** discrete, Number of extra tutoring or coaching sessions attended per week.
- **Sleep_Hours: continuous,** Average hours of sleep per day.
- **Previous_Scores:** continuous, Past exam results.
- **Physical_Activity:** discrete, Number of exercises and physical activity sessions per week.
- **Exam_Score:** numeric, Final exam score or academic performance (target variable).

Note: The **Exam_Score** is not a predictor; it is the **dependent (target) variable**, but it still belongs to the quantitative category.

And there are 13 qualitative predictor variables (ordinal, nominal) in the examined dataset, which are:

- **Parental_Involvement:** ordinal, Level of parents' participation in the student's education (Low, Medium, High).
- **Access_to_resources:** ordinal, Availability of learning materials and digital tools (Low, Medium, High).
- **Extracurricular_Activities:** nominal, whether the student participates in extracurricular activities (Yes/No).
- **Motivation_Level:** ordinal, The student's motivation level toward studies (Low, Medium, High)
- **Internet_Access:** nominal, Does the student have internet access at home (Yes/No)
- **Family_Income:** ordinal, Economic status of the student's family (Low, Medium, High).
- **Teacher_Quality:** ordinal, Quality of instruction received from teachers (Low, Medium, High).
- **School_Type:** nominal, type of educational institution (Public/Private).
- **Peer_Influence:** ordinal, Nature of peer influence (Positive, Neutral, Negative).
- **Learning_Disabilities:** nominal, indicates if the student has a learning disability (Yes/No).
- **Parental_Education_Level:** ordinal, Highest education level of parents (High School, College, Postgraduate)
- **Distance_from_Home:** ordinal, Distance between student's home and school (Near, Moderate, Far)
- **Gender:** nominal, Student's gender (Male/Female).

Together, these qualitative and quantitative variables provide a comprehensive profile of the social, academic, environmental, and personal factors that may contribute to variations in students' academic achievement.

Why is regression an appropriate method?

Student academic performance is shaped by a wide range of personal, educational, and social factors, which in this dataset are represented by multiple predictor variables such as study habits, parental involvement, school environment, motivation, sleep duration, and access to learning resources. Because performance outcomes are not determined by a single factor but rather by the combined influence of many conditions, a statistical approach capable of handling multiple predictors at once is required. Regression analysis provides this framework by allowing the relationship between the final exam score (the response variable) and numerous explanatory variables to be quantified and assessed simultaneously. Through this method, it becomes possible not only to predict student achievement based on observed characteristics but also to evaluate which variables have the strongest or weakest effect, and whether those effects are statistically significant. Given that the dataset contains a continuous outcome measure (final exam score) along with several quantitative and qualitative predictors, it aligns directly with the requirements of a multiple linear regression model, making this technique an appropriate and effective tool for

examining determinants of academic performance and supporting evidence-based interpretation and prediction.

Methodology-Literature Review

Literature Review

The STAT 2240 Student Manual provides a theoretical foundation for understanding how Relationships between variables are assessed and modeled using regression analysis. According to the manual, regression is a statistical technique used to identify, quantify, and model the relationship between a response variable and one or more explanatory variables. This process begins with the visual tools, such as scatter plots, which help determine whether a linear relationship may exist between pairs of variables.

As stated in Chapter 1 Student Manual (page 5 - 32), scatter plots reveal the direction, shape, and strength of associations, whereas correlation analysis offers a numerical measure of the linear link. Because scatter plots alone cannot quantify strength or account for the impacts of numerous elements concurrently, the manual introduces correlation and regression as key tools for further research.

The manual distinguishes between deterministic models in Student Manual Chapter 4 (pages 102 - 121), which assume no random variability, and statistical models, which use a random error term (ϵ) to account for unexplained variance. The distinction is critical in regression analysis because random error influences how we estimate parameters, generate predictions, and do inference.

Overall, the Student Manual provides a conceptual framework for studying real-world datasets: observe associations visually, quantify them using correlation, and finally evaluate them using multiple linear regression while ensuring that all model assumptions are met.

Methodology

The methodology employed in this project is entirely aligned with the processes specified in the STAT 2240 Student Manual Chapter 1 (pages 5 - 32). The first stage involves preliminary data exploration, where scatter plots and correlation coefficients are used to examine pairwise relationships among variables. Scatterplots offer a visual sense of whether a linear pattern exists, whereas correlation coefficients measure the degree and direction of these correlations. The correlation matrix serves as a valuable tool for screening variables and identifying possible multicollinearity, ensuring that the predictors used for the model are adequate and that strong linear correlations between explanatory variables are discovered early. This manual defines the general multiple linear regression model as:

$$Y = \beta_0 + \beta_1\chi_1 + \beta_2\chi_2 + \cdots + \beta_k\chi_k + \epsilon .$$

This model has linear parameters and satisfies one of the main deterministic assumptions (MLR1). In the STAT 2240 Student Manual, Chapter 2 (pages 33 - 68), the coefficient estimates will be calculated using Ordinary Least Squares (OLS), which will minimize the Residual Sum of Squares (RSS). According to the manual, regression modelling enables us to analyze the contribution of each explanatory variable, evaluate the model's overall performance using the F-tests, and determine the significance of individual predictors using t-tests. Measures of fit, including R^2 , *Adjusted* R^2 , further assessment of the model's ability to explain response

variability. Although these computations will be performed in subsequent stages of the project, the methodological foundation is laid here.

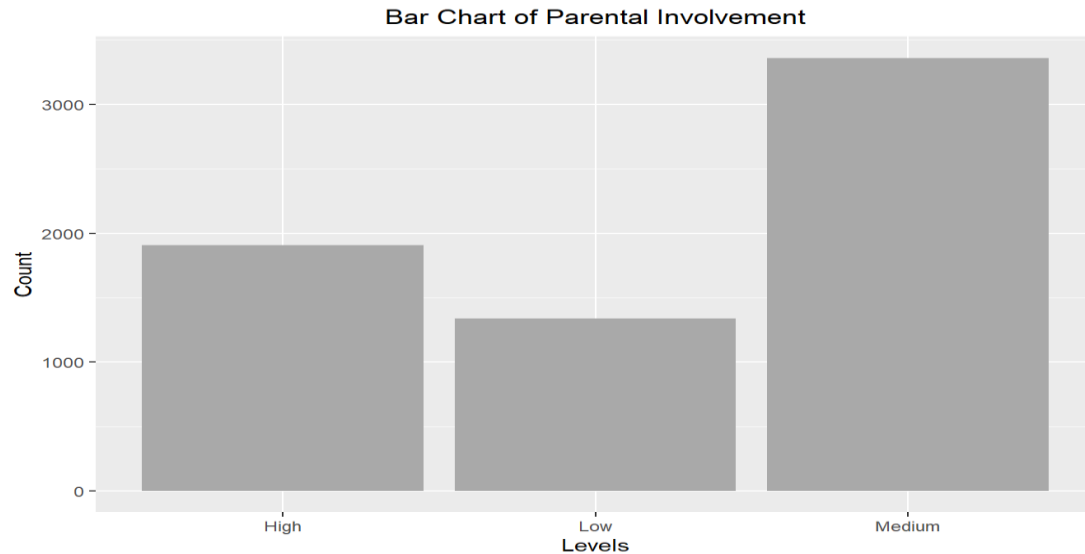
Before fitting the model, the Student Manual Chapter 3 (pages 76 - 102) emphasizes the significance of testing the model's deterministic assumptions. These include the model's parameters are linear (MLR1) and accurately defined (ML2), which means all important variables are eliminated, and the functional form is suitable. Reliable estimation requires the absence of multicollinearity. Strong linear correlations among predictors diminish the invertibility of the matrix $X'X$ and raise the variance of coefficient estimations. The correlation matrix and scatter plots help to discover such issues, and the manual recommends remedial steps such as removing redundant variables or revising the model structure if multicollinearity is present. Once the model is estimated, the Student Manual provides many assumptions linked to the random elements of the error term that must be checked to guarantee proper inference. Residuals should have a mean of zero, constant variance, independence, and a nearly normal distribution. Residual-based diagnostic plots, such as the residuals ~fitted values plot to verify the mean and variance, residual order plots to validate independence, Q-Q plots, and histograms to measure normality, are used to investigate the assumptions. Because outliers or high-leverage points can skew model estimates and jeopardize inference, it is also necessary to identify the influential observations using methods like Cook's Distance. The manual suggests fixes including logarithmic or square-root transformations, eliminating significant points, or re-specifying the model if any of these presumptions are broken.

Lastly, the Student Manual Chapter 4 (pages 102- 121) + Additional slides provided clarify that choosing the best model frequently necessitates a methodical variable selection process. Methods such as forward selection, backward elimination, and stepwise selection

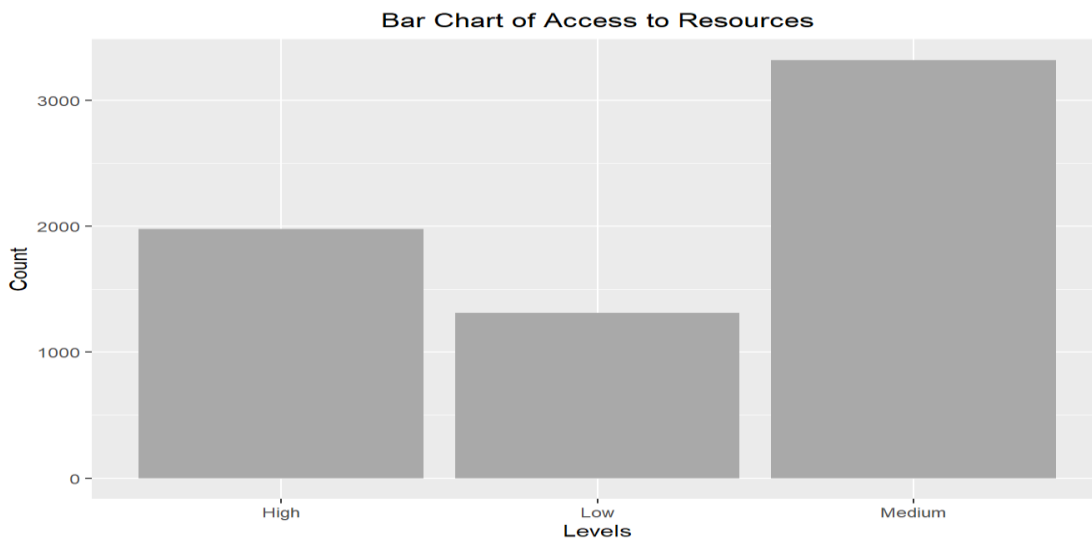
analyze the contribution of variables and assist in developing a parsimonious model that balances complexity and explanatory power. These methods, which will be used in Phase 2, direct the regression model's improvement. The process shown here, which starts with exploratory analysis and moves through the model design, assumption verification, and model selection, generally mirrors the whole modelling framework detailed in the STAT 2240 Student Manual.

Descriptive Statistics

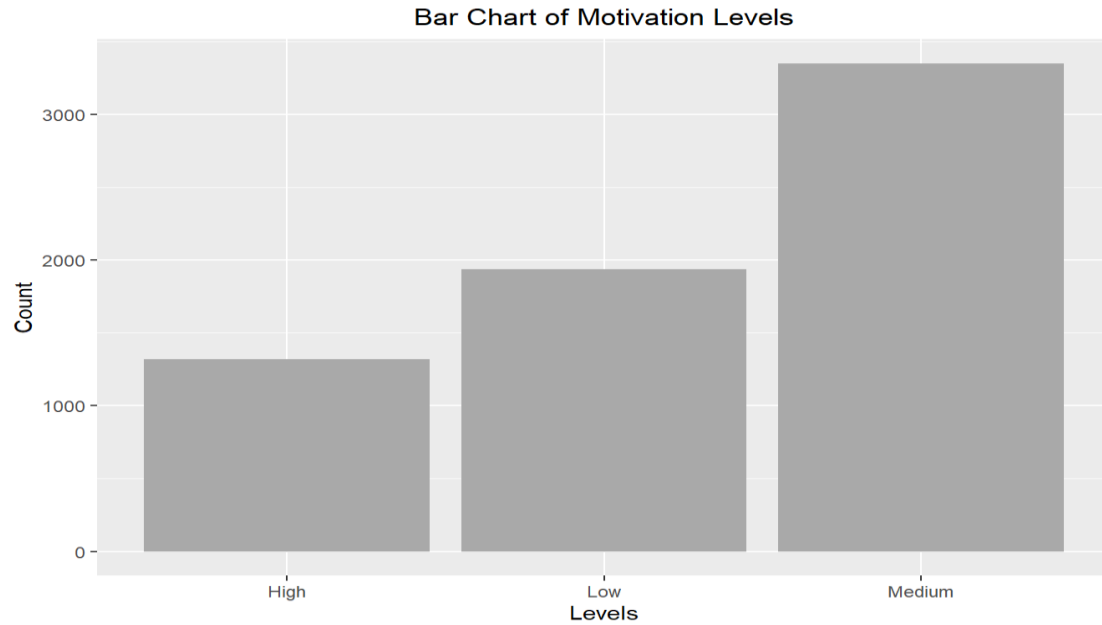
Qualitative Variables



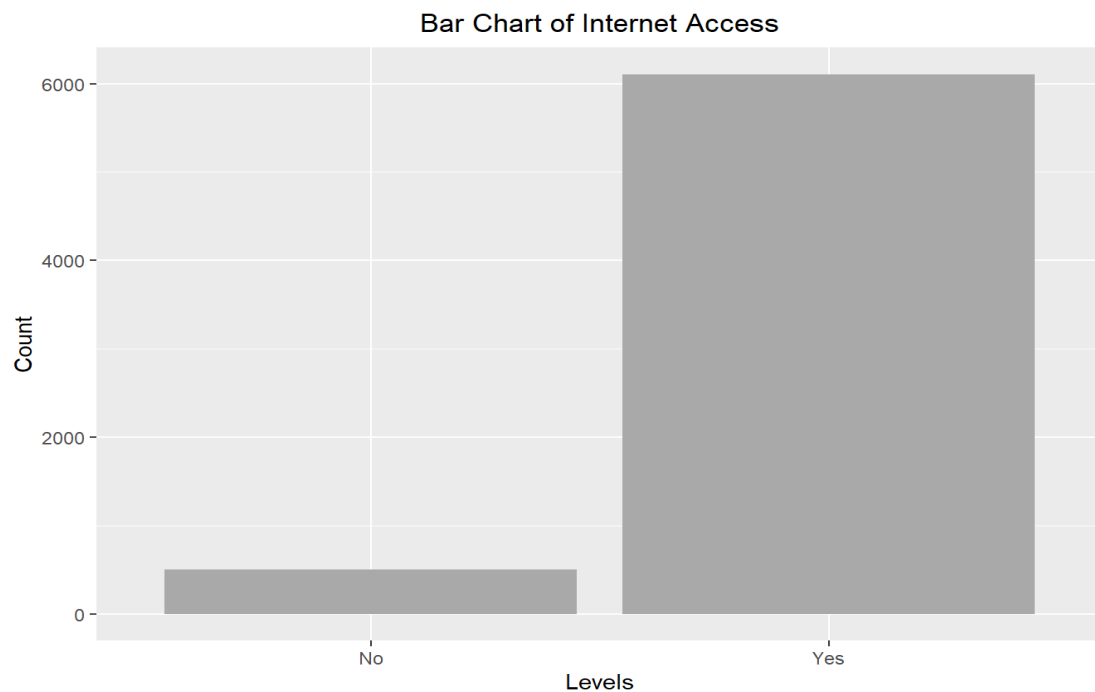
For **Parental_Involvement**, “Medium” is the most frequent category, followed by “High.” This indicates that most students receive at least a moderate level of support from their parents



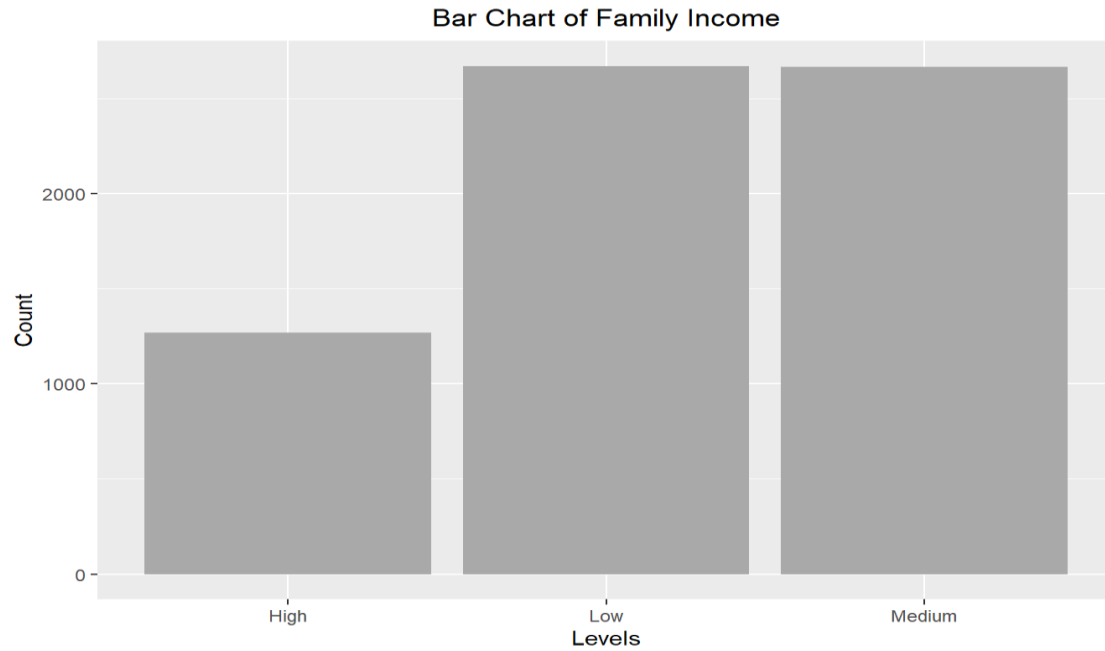
This chart shows a very similar pattern to parental involvement, with the "Medium" level of resource access being the most common.



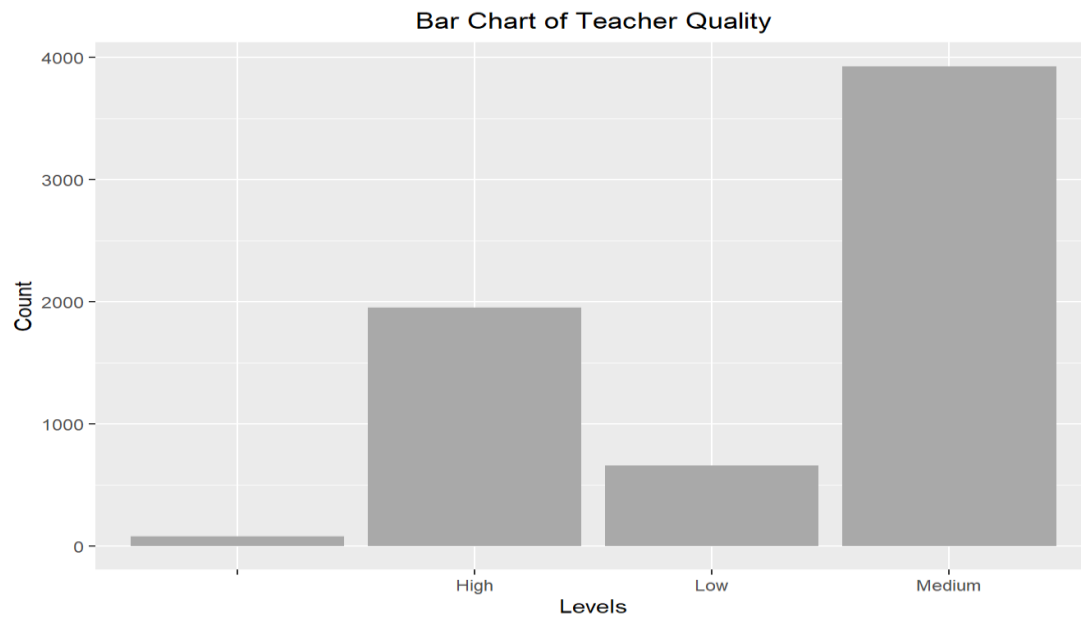
This chart indicates that "Medium" is the predominant motivation level among students, with roughly equal but smaller groups reporting high and low motivation.



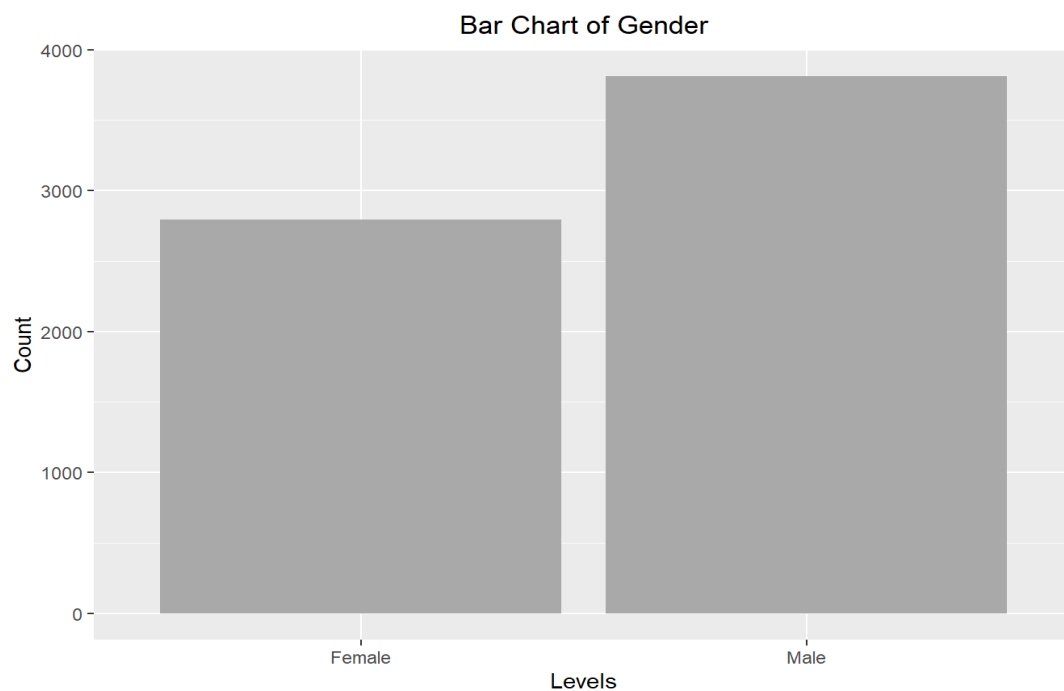
This chart presents a dramatic imbalance, with an overwhelming majority of students having internet access ("Yes").



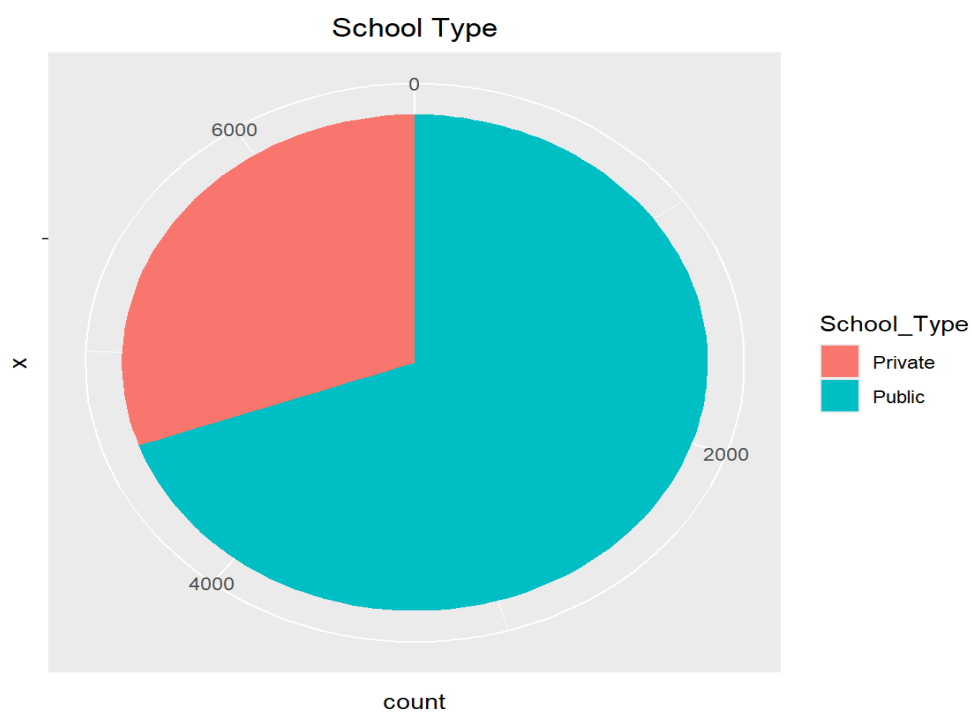
This chart reveals that "Low" income is the single most frequent category, closely followed by "Medium."



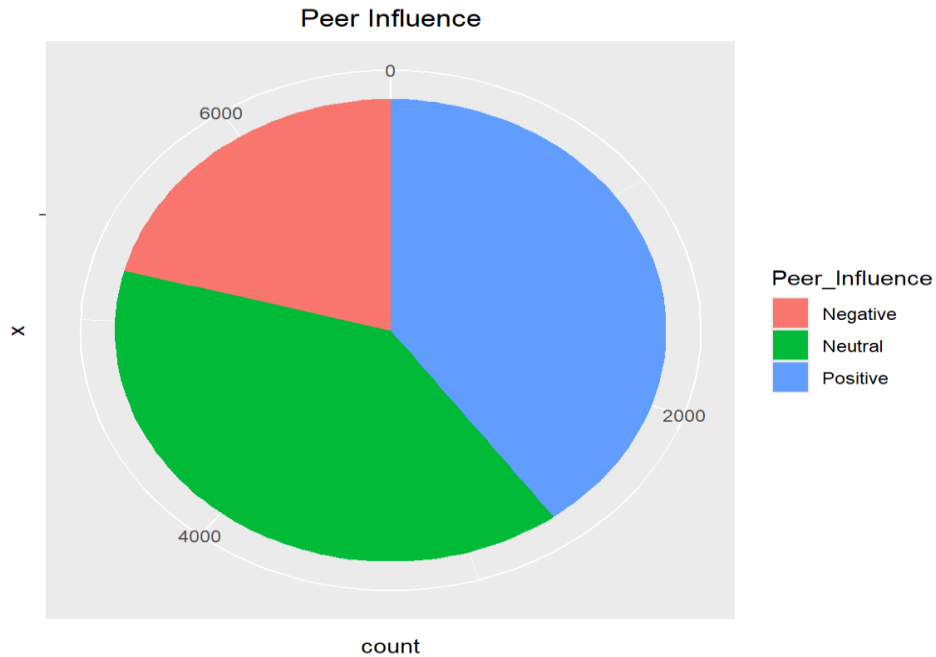
This chart shows that "Medium" is the most frequent rating for teacher quality.



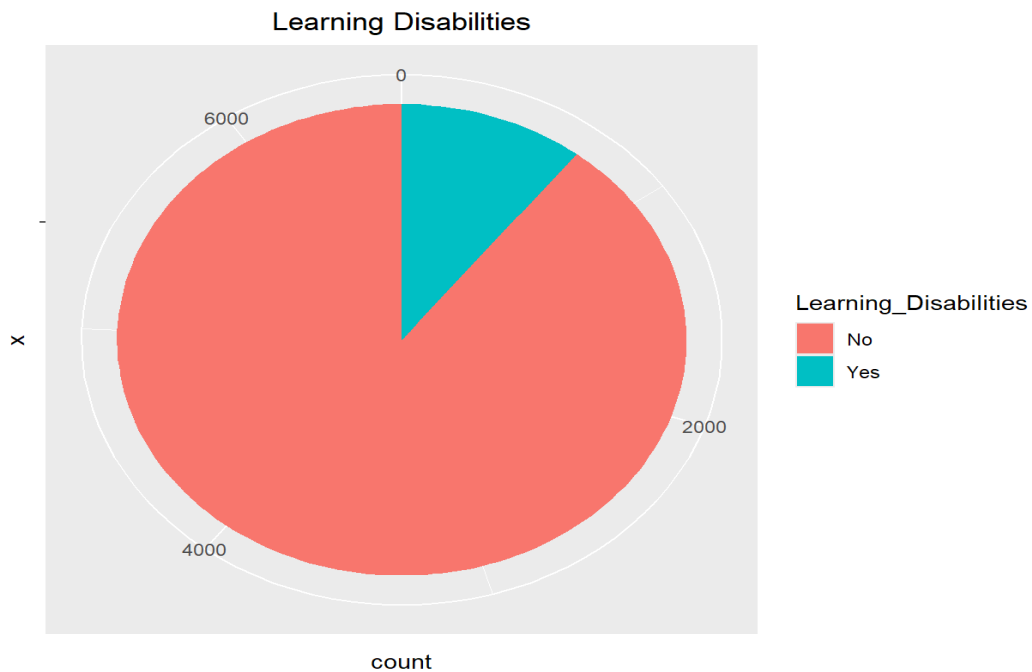
This chart shows a simple majority of male students over female students.



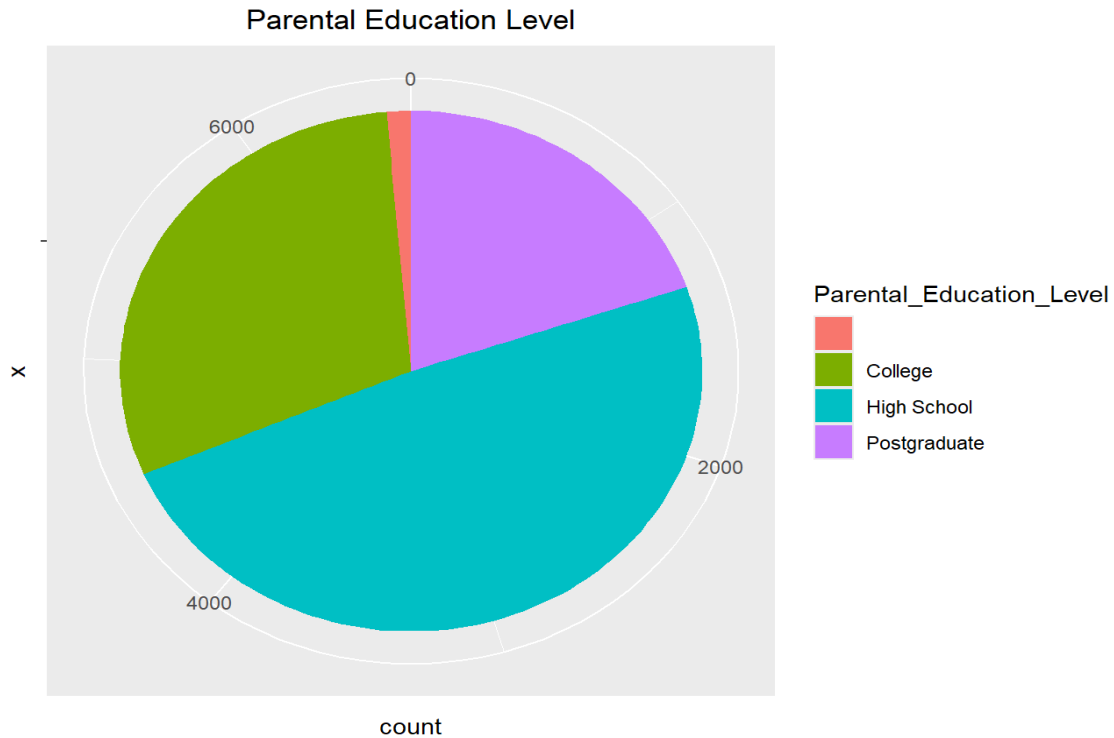
This figure shows that the experiences and outcomes in this study are predominantly shaped by the public school environment.



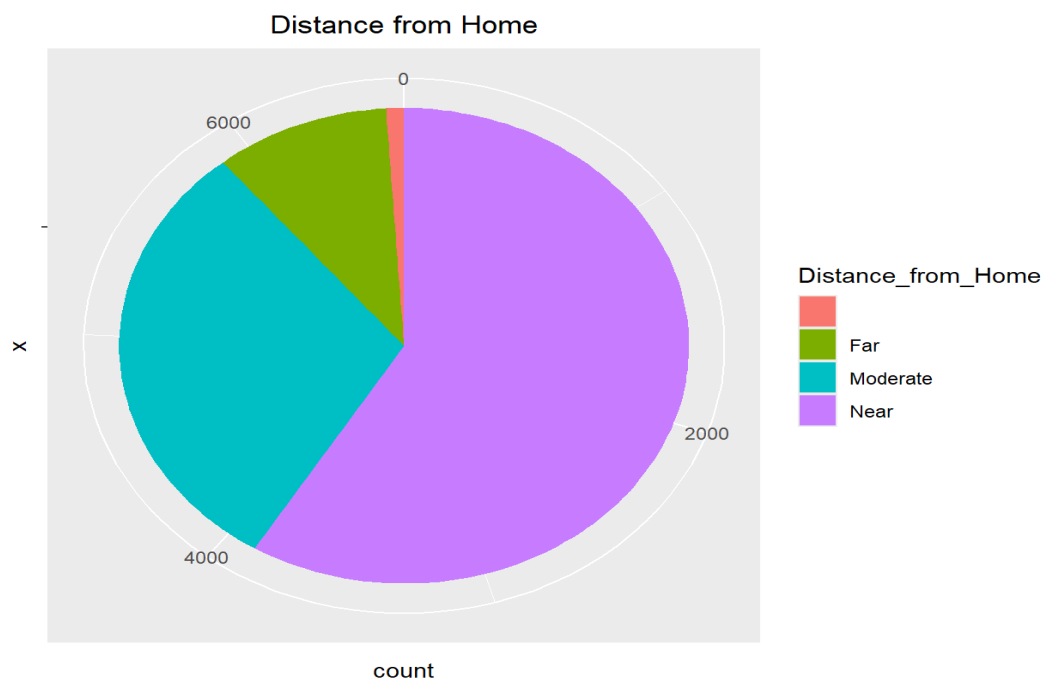
This figure indicates that overtly negative peer pressure is less common than positive or neutral social influences among students.



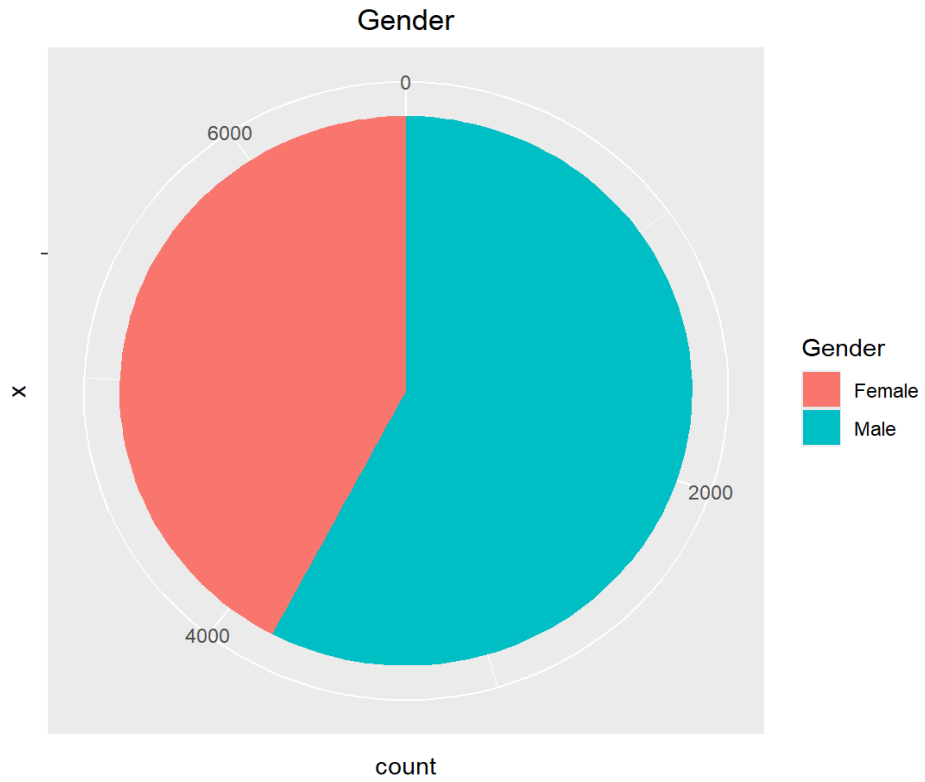
This figure confirms that students with identified learning disabilities represent a clear minority within the studied population.



This figure indicates that a high school diploma is the most common highest educational attainment among parents, suggesting varied but often non-advanced academic backgrounds at home.



This figure shows that a long commute is not a typical stressor or time constraint for the vast majority of these students.



This figure indicates a modest overrepresentation of male students in the sample.

Frequency and Mode Tables

Frequency and Mode tables were generated for the categorical variables, showing the exact counts and percentages for each category. These tables confirm the same patterns observed in the bar and pie charts.

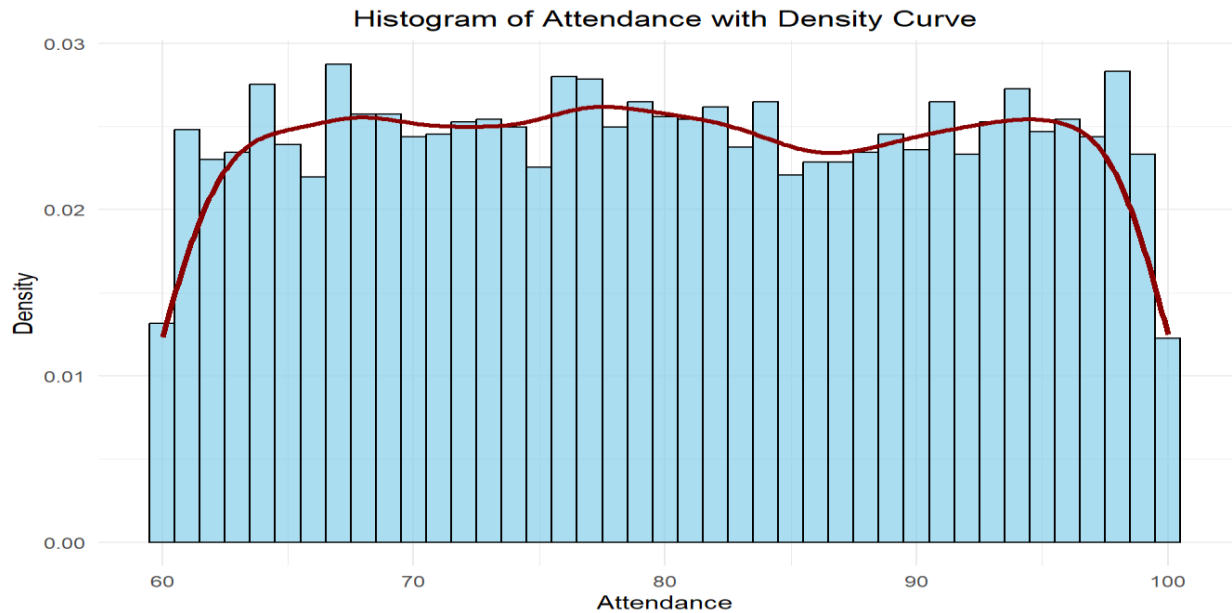
Mode Table

VARIABLES	MODE
Parental Involvement	Medium
Access to Resources	Medium
Extracurricular Activities	Yes
Motivation Level	Medium
Internet Access	Yes
Family Income	Low
Teacher Quality	Medium
School Type	Public
Peer Influence	Positive
Learning Disabilities	No
Parental Education Level	High School
Distance from Home	Near
Gender	Male

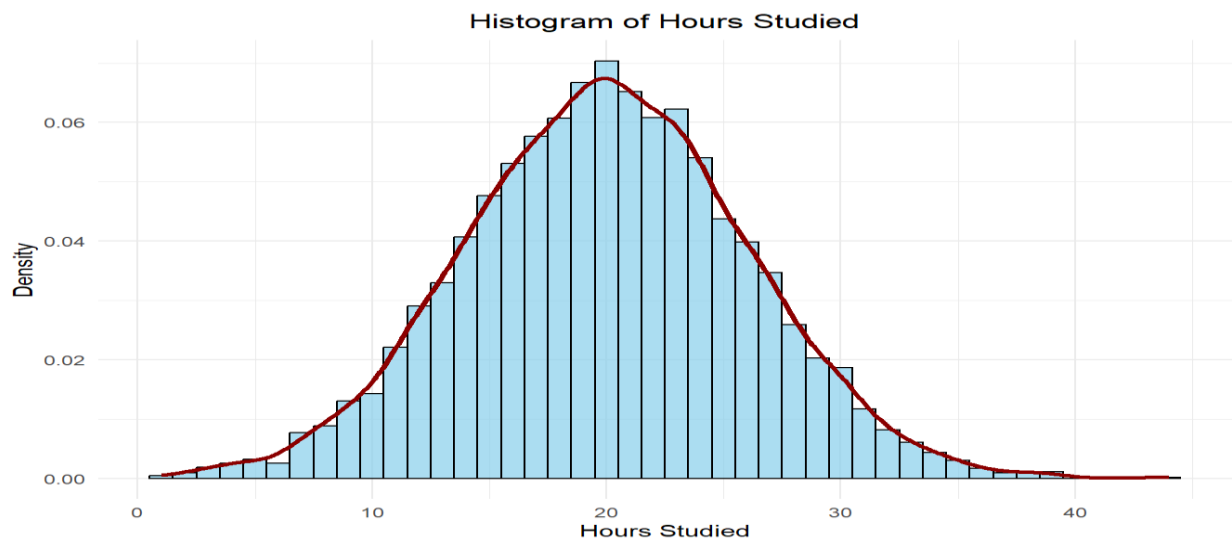
FREQUENCY TABLE

VARIABLE	CATEGORY	FREQUENCY	RELATIVE FREQ
Parental Involvement	High	1908	28.9
	Low	1337	20.3
	Medium	3362	50.9
Access to Resources	High	1975	29.9
	Low	1313	19.9
	Medium	3319	50.2
Extracurricular Activities	No	2669	40.4
	Yes	3938	59.6
Motivation Level	High	1319	20.0
	Low	1937	29.3
	Medium	3351	50.7
Internet Access	No	499	7.6
	Yes	6108	92.4
Family Income	High	1269	19.2
	Low	2672	40.4
	Medium	2666	40.3
Teacher Quality	(Missing)	78	1.2
	High	1947	29.5
	Low	657	9.9
	Medium	3925	59.4
School Type	Private	2009	30.4
	Public	4598	69.6
Peer Influence	Negative	1377	20.8
	Neutral	2592	39.2
	Positive	2638	39.9
Learning Disabilities	No	5912	89.5
	Yes	695	10.5
Parental Education Level	(Missing)	90	1.4
	College	1989	30.1
	High School	3223	48.8
	Postgraduate	1305	19.8
Distance from Home	(Missing)	67	1.0
	Far	658	10.0
	Moderate	1998	30.2
	Near	3884	58.8
Gender	Female	2793	42.3
	Male	3814	57.7

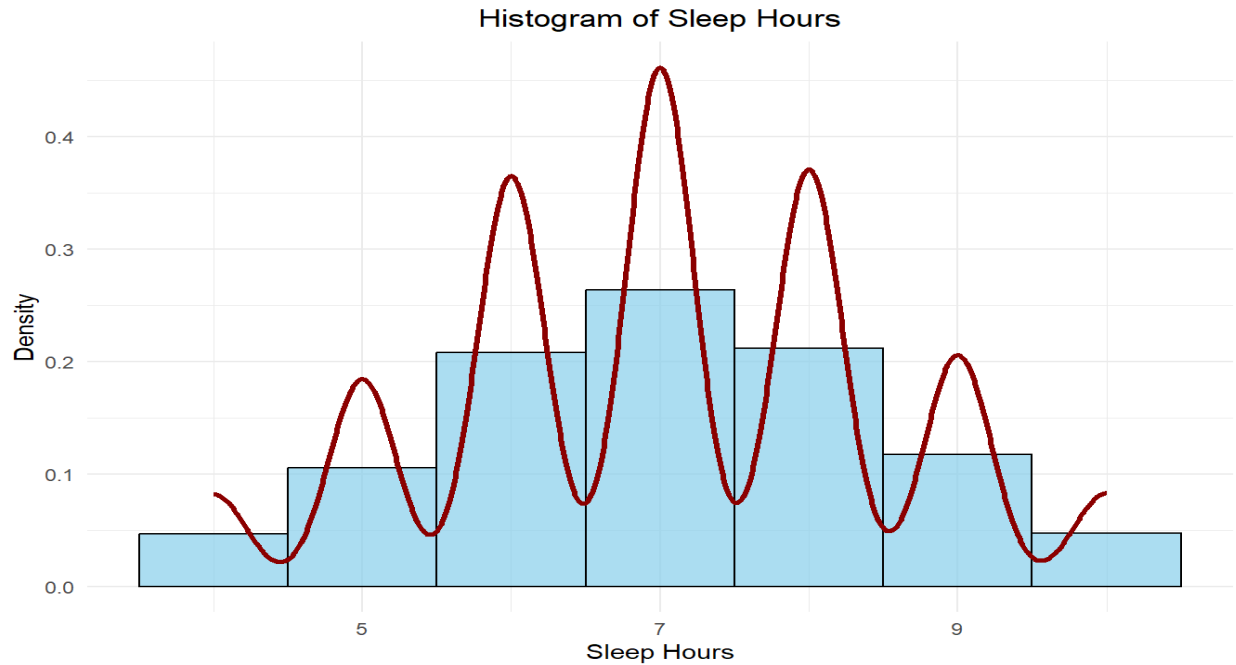
Quantitative Variables



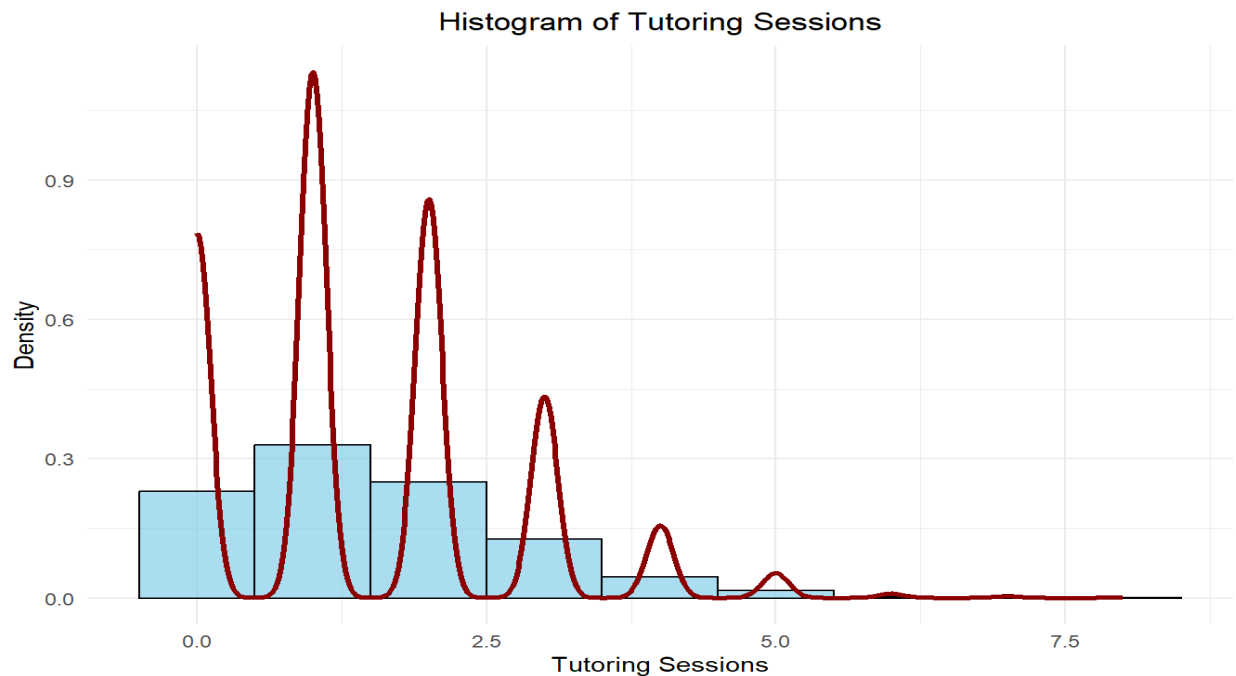
This figure shows a roughly symmetric, bell-shaped distribution centered around 80%, which indicates that most students have good and consistent attendance, with few extreme cases of very low or perfect attendance.



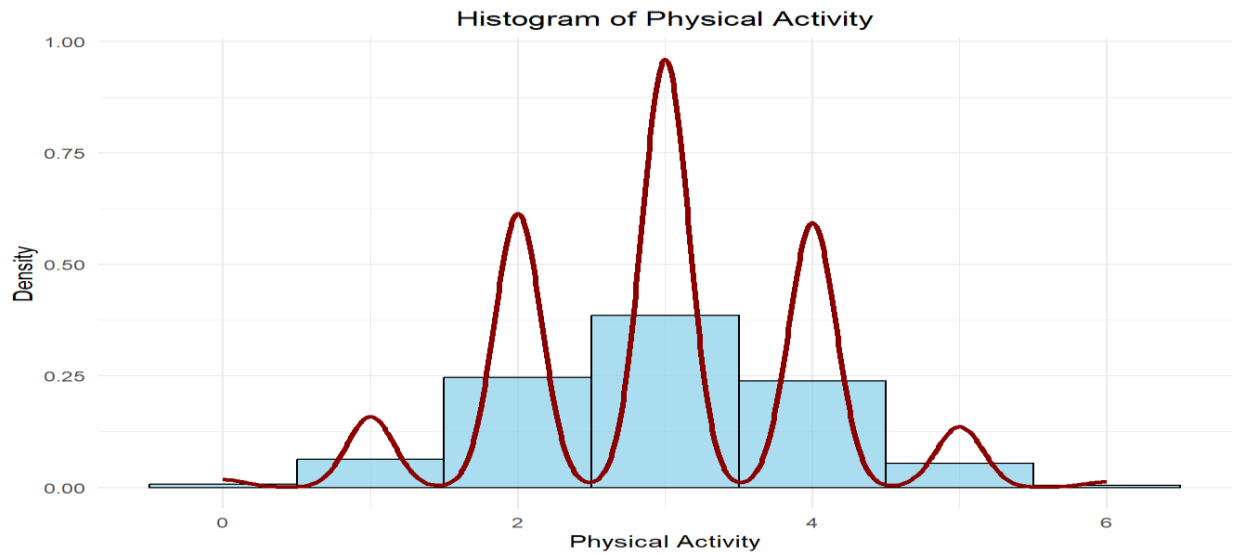
This figure shows a symmetric distribution with a peak near 20 hours, suggesting that while a typical student studies around 20 hours, a notable portion of students study significantly more, indicating a high-achieving or highly motivated subgroup.



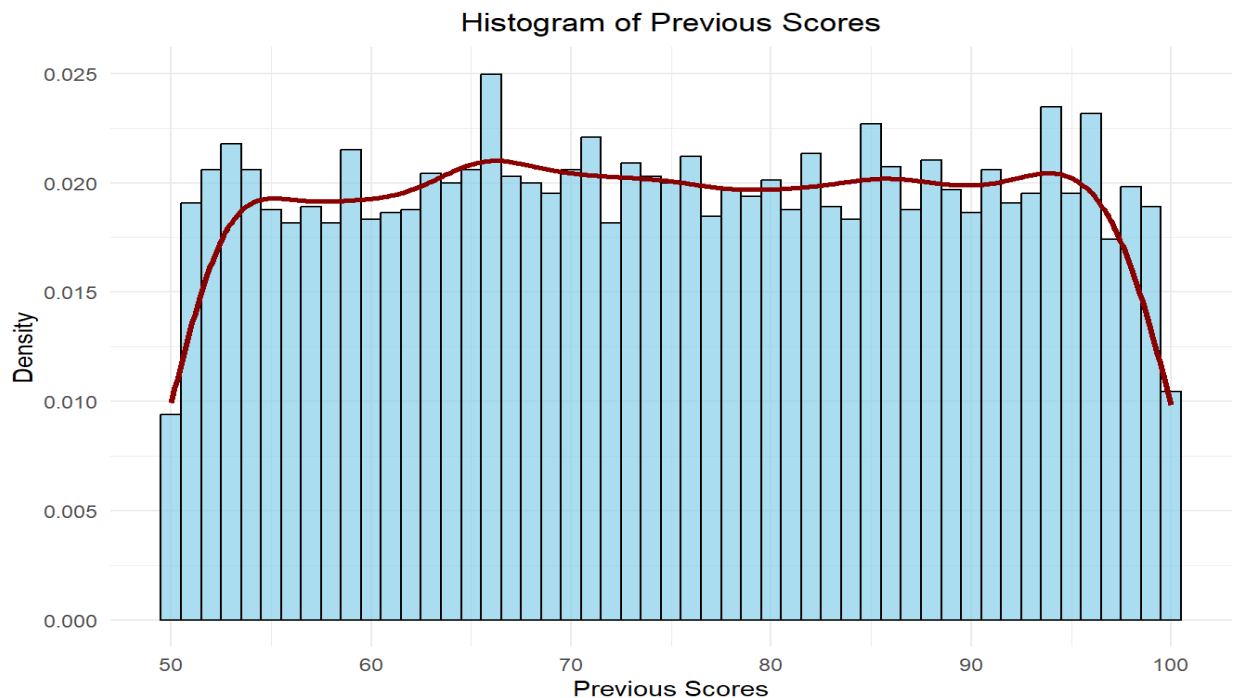
This figure shows a symmetric, tightly clustered distribution around 7 hours, indicating that sleep habits are very consistent across the student population, with most meeting the recommended nightly amount.



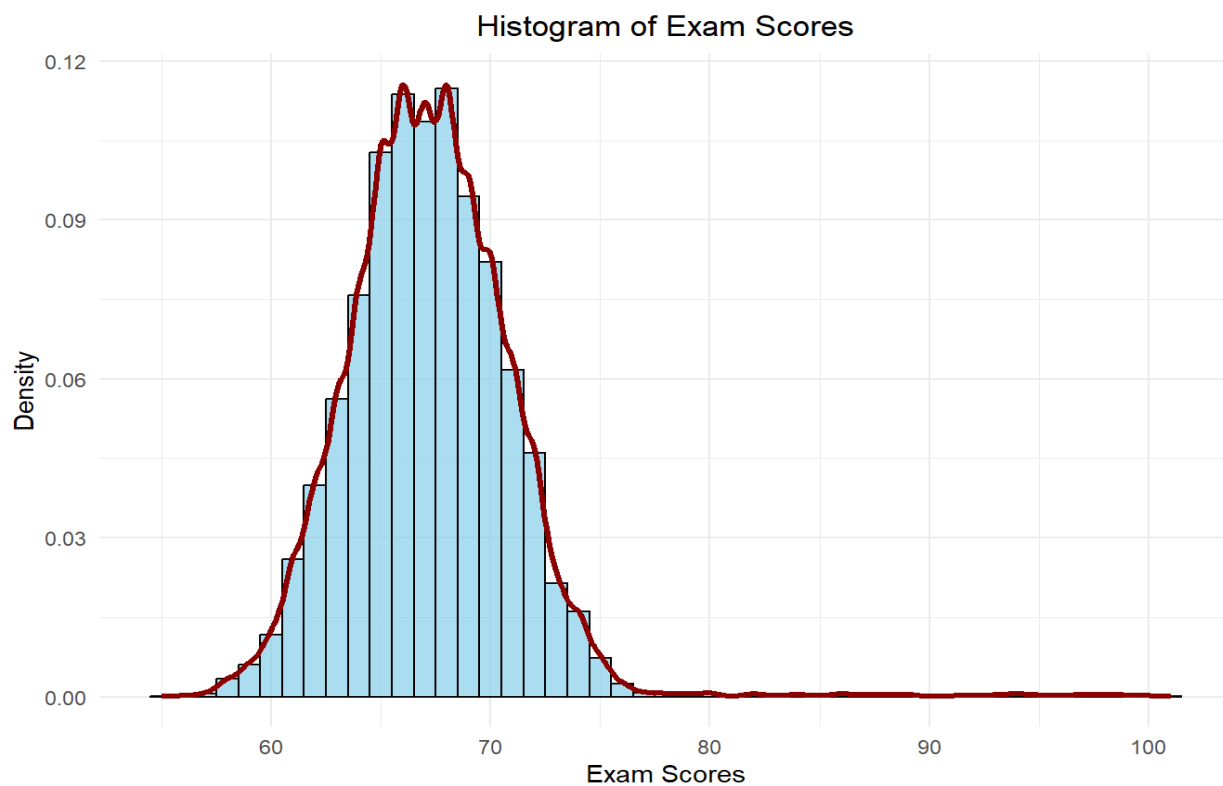
This figure shows a strongly right-skewed distribution where most students attend 0–2 sessions. This reveals that tutoring is not a common practice for the majority and is utilised heavily by only a small subset of students, likely those seeking extra help.



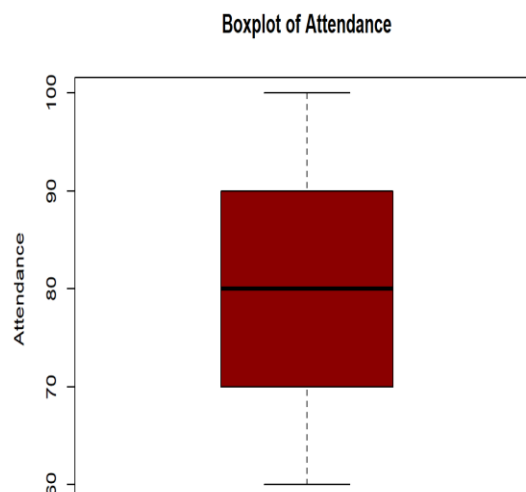
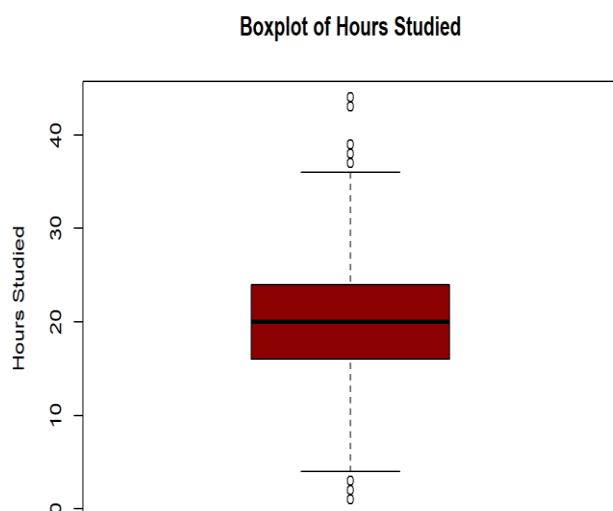
This figure shows a symmetric distribution, clearly indicating that low levels of physical activity are the norm for this student population, with very few students engaging in high levels of exercise.



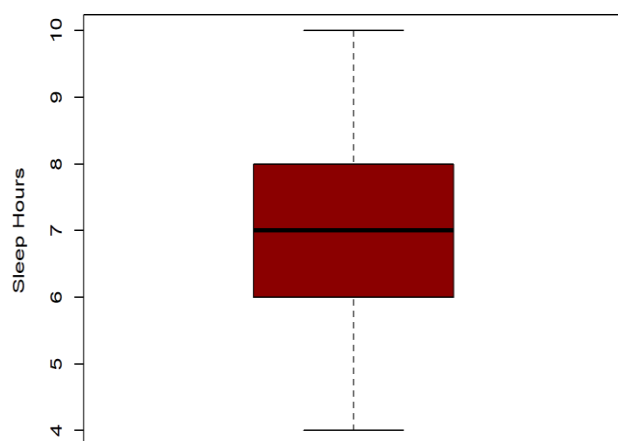
This figure shows a symmetric, normal distribution centered around 75, which suggests that prior academic performance in this group is varied but follows a predictable pattern, with most students performing on average well.



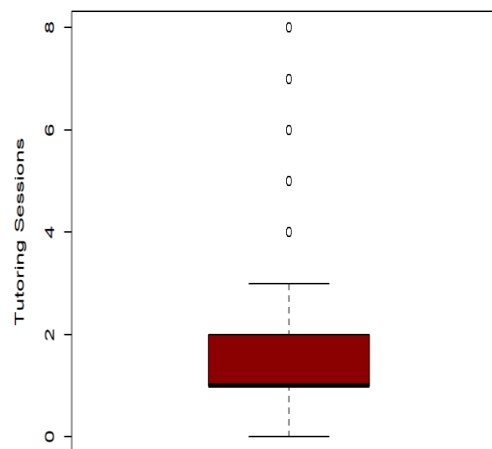
This figure shows a symmetric, normal distribution with a mean near 67, which is lower and tighter than the previous scores. This indicates that the final exam was challenging, resulting in scores that are generally lower and less variable than students' past performance.



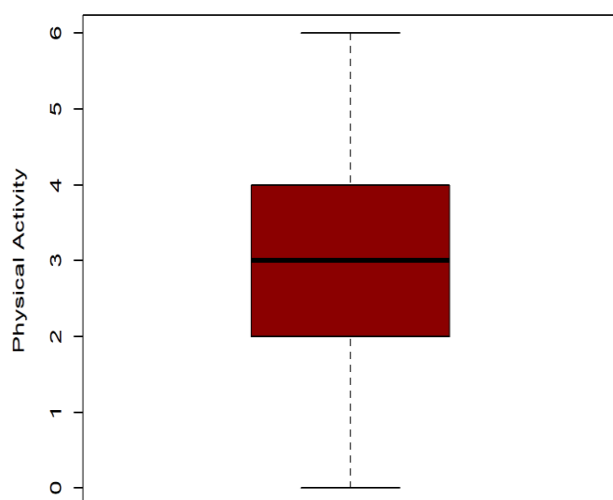
Boxplot of Sleep Hours



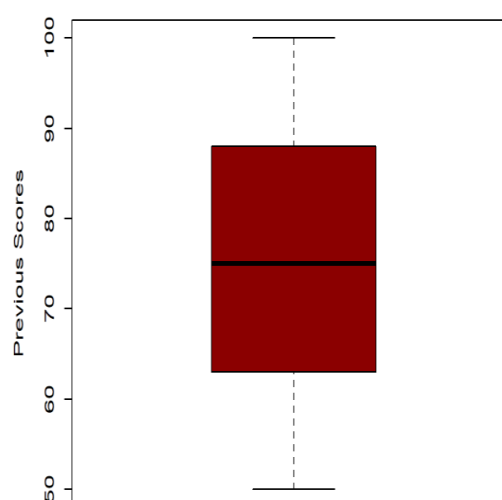
Boxplot of Tutoring Sessions

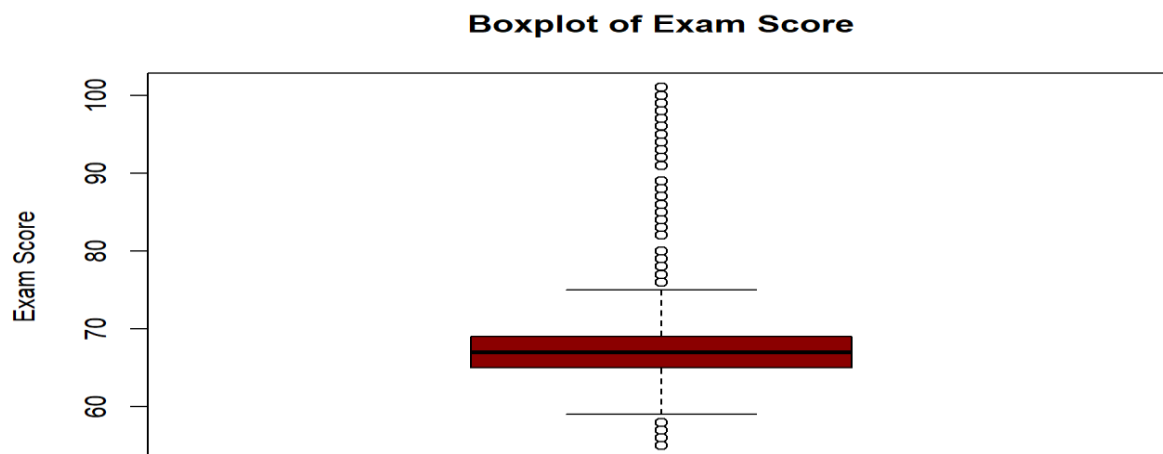


Boxplot of Physical Activity



Boxplot of Previous Scores



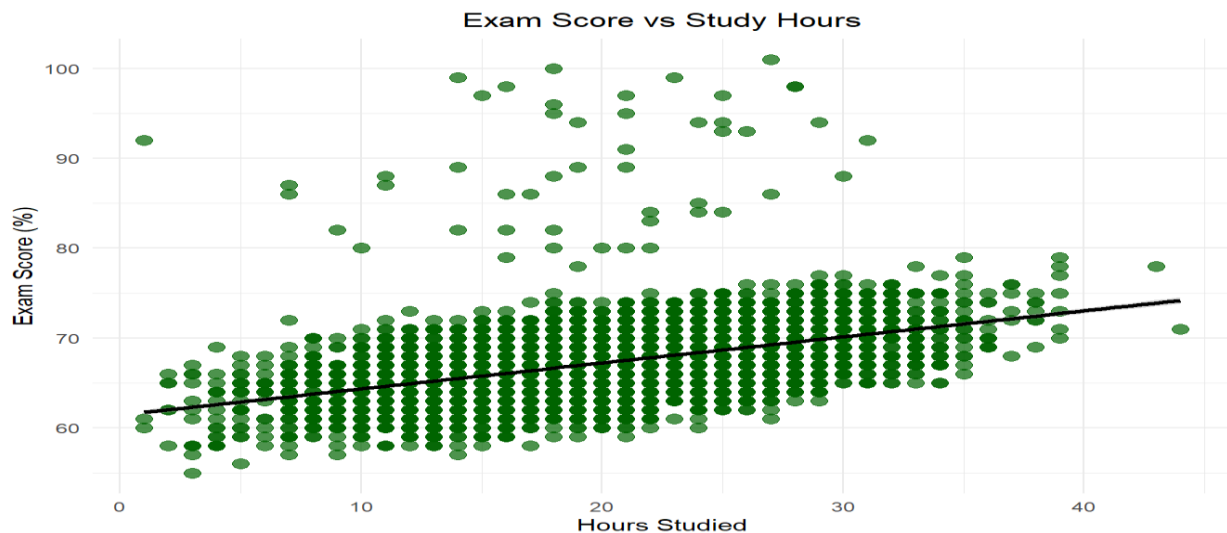


In the search for outliers, there were only three variables with outliers: Hours studied, tutoring sessions, and exam scores.

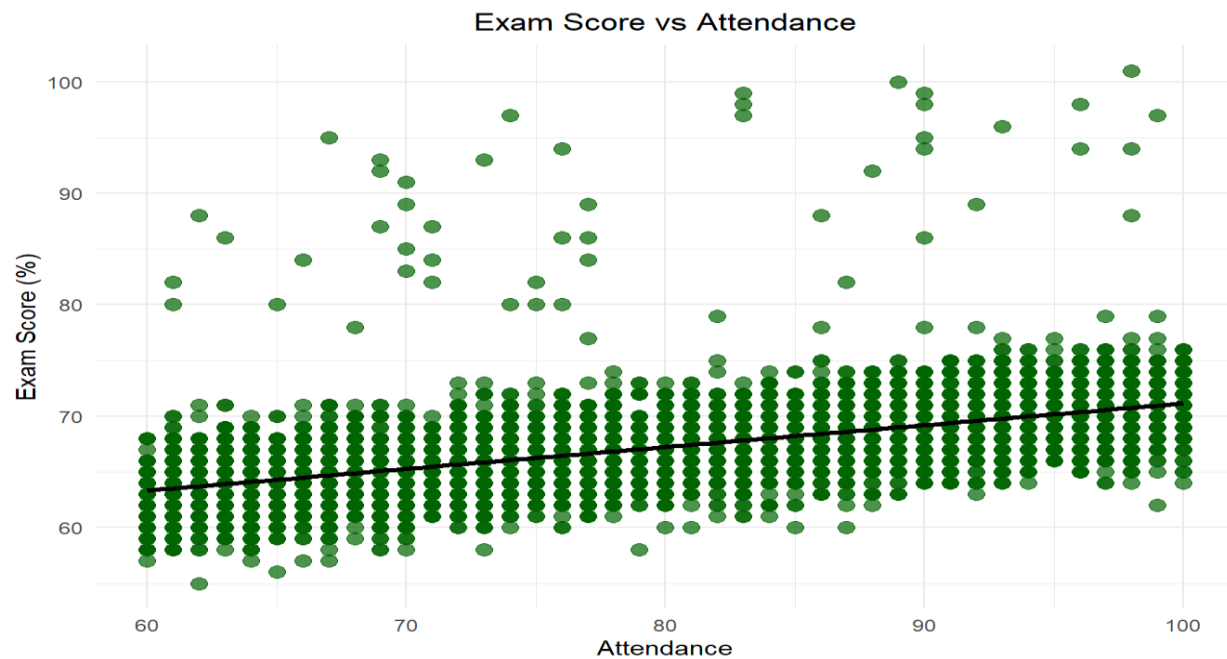
Summary Table of Quantitative Variables

A summary table was generated for the quantitative variables displaying the mean, median, minimum, maximum, variance, standard deviation, and interquartile range. The table confirms the pattern observed in both the histograms and boxplots.

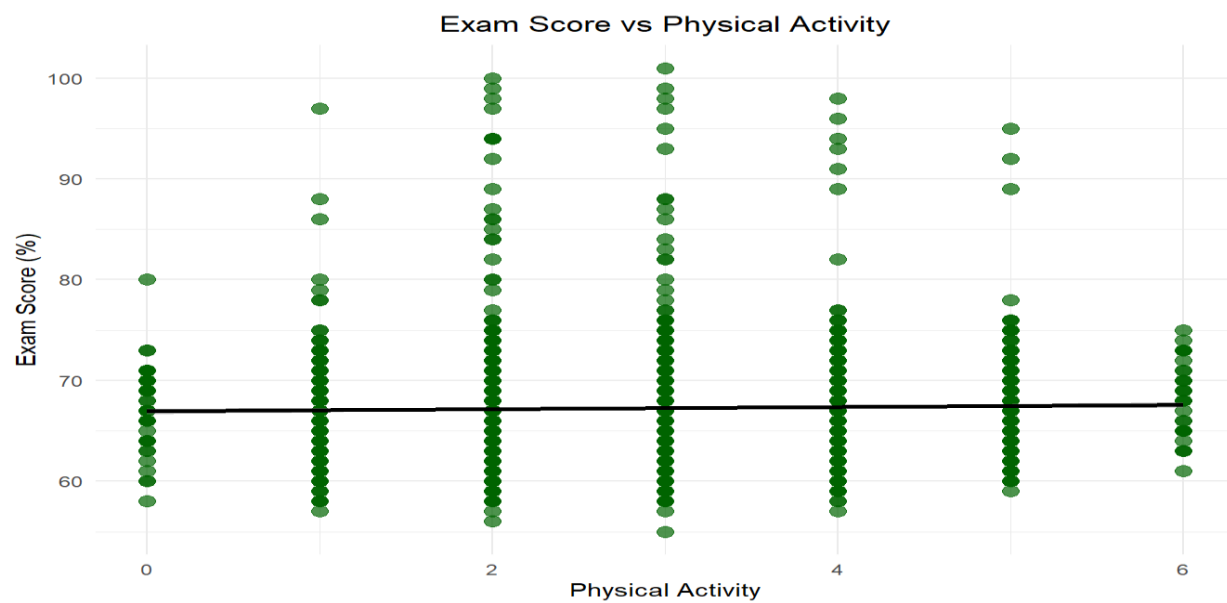
VARIABLE	MEAN	MEDIAN	MIN	MAX	VAR	SD	IQR
Hours Studied	19.98	20	1	44	35.89	5.99	8
Attendance	79.98	80	60	100	133.34	11.55	20
Sleep Hours	7.03	7	4	10	2.16	1.47	2
Previous Scores	75.07	75	50	100	207.35	14.40	25
Tutoring Sessions	1.49	1	0	8	1.51	1.23	1
Physical Activity	2.97	3	0	6	1.06	1.03	2
Exam Score	67.24	67	55	101	15.13	3.89	4



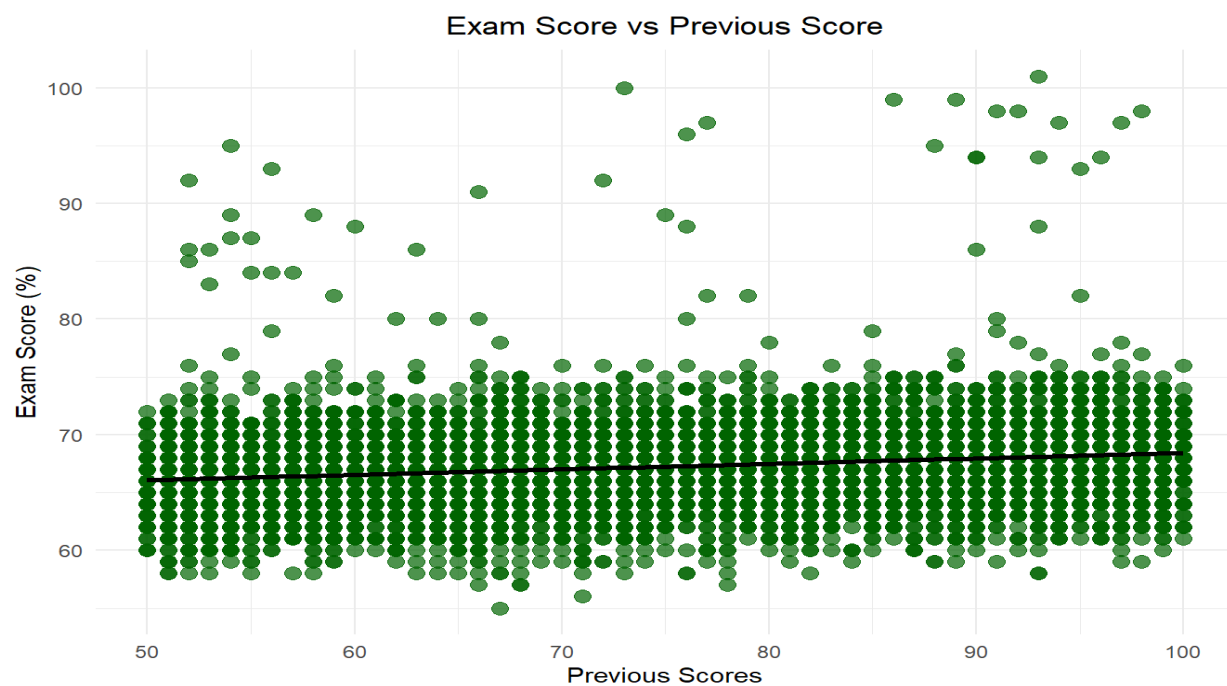
A moderate positive linear relationship is observed. As study hours increase, exam scores tend to increase, but the relationship is neither strong nor weak across all students.



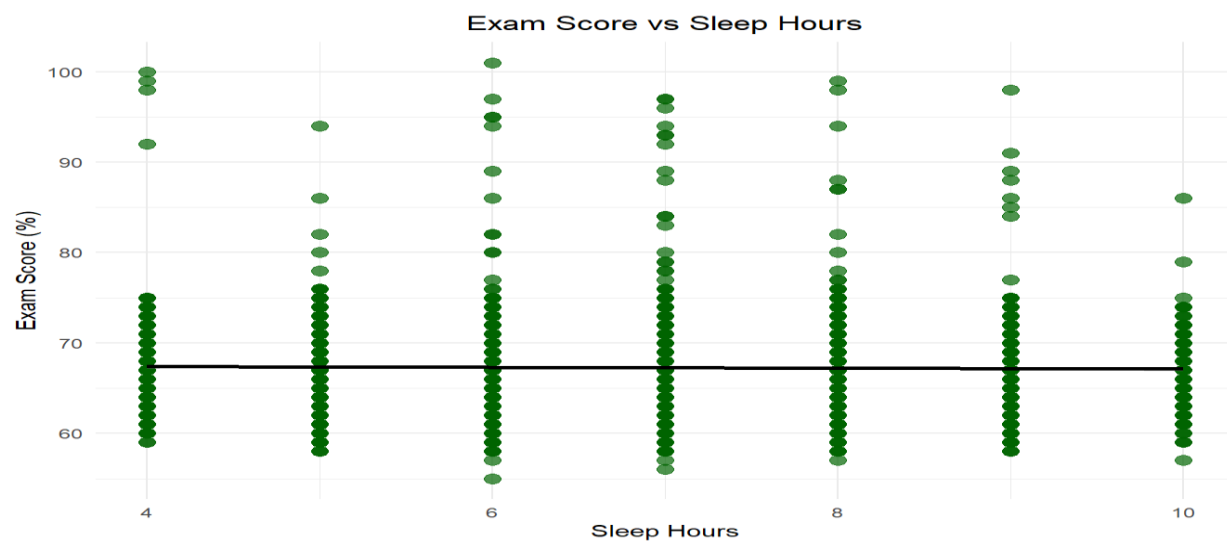
Just as in study hours, the relationship is a moderate linear one.



No clear linear relationship is observed. Physical activity levels do not appear to linearly affect exam scores in this dataset.



An extremely slight positive linear relationship is observed. Higher previous scores are generally associated with higher final exam scores, indicating that prior academic performance is a relevant predictor.



A weak, basically non-existent negative linear relationship is observed.



There is not much of a relationship.

Correlation Matrix of all Variables

	Hours Studied	Attendance	Sleep Hours	Previous Score	Tutoring Sessions	Physical Activity	Exam Score
Hours studied	1	-0.0099	0.0109	0.0248	-0.01428	0.0046	0.4454
Attendance	-0.0099	1	-0.01591	-0.0202	0.01432	-0.0224	0.5811
Sleep Hours	0.0109	-0.0159	1	-0.0218	-0.0122	-0.0038	-0.0170
Previous Scores	0.0248	-0.2018	-0.0217	1	-0.0131	-0.0113	0.1751
Tutoring Sessions	-0.0143	0.01432	-0.0122	-0.0131	1	0.01773	0.1565
Physical Activity	0.0046	-0.2243	-0.0004	-0.01127	0.01773	1	0.2782
Exam Score	0.4454	0.5810	-0.0170	0.1751	0.1565	0.0278	1

Observation from the Correlation Matrix

The correlation matrix reveals the strength and direction of linear relationships between all quantitative variables and the exam score. Key observations include:

Strongest Positive Correlations with Exam Score:

- Attendance (0.5811): The strongest predictor, indicating that higher attendance is strongly associated with higher exam scores.
- Hours Studied (0.4454): A moderately strong predictor, reinforcing the importance of study time.

Weaker but Notable Positive Correlations:

- Physical Activity (0.2782)

- Previous Scores (0.1751)
- Tutoring Sessions (0.1565)

Negligible/Negative Correlations:

- Sleep Hours (-0.0170): Essentially no linear relationship with exam scores.

Suggestion Based on Observations

Given these observations, the following steps are recommended for the next phase of the regression analysis:

Proceed with a multiple linear regression model using Attendance and Hours Studied as the primary explanatory variables, as they show the strongest individual relationships with the exam score. Previous Scores should also be included due to their moderate correlation and theoretical importance.

Variables like Tutoring Sessions and Physical Activity could be tested in the model but may be candidates for removal if they are found to be insignificant during model refinement (e.g., using stepwise selection or p-value evaluation). Sleep Hours can likely be excluded from the initial model due to their near-zero correlations.

Inferential Statistics

Step 1: Multicollinearity Diagnostics

The first step outlined in the student manual was to check for collinearity in our independent variables. Multicollinearity was assessed using the Generalised VIF. The GVIF and adjusted GVIF values were approximately close to 1, which indicates a complete absence of multicollinearity amongst the independent variables. (See figure 1.1). As a result, all predictors were kept for inferential studies. Alternatively, the correlation matrix for the continuous predictors shows very weak pairwise correlation, with the coefficients ranging between -0.02 and 0.02. Since the values are close to zero, we reached the conclusion that the predictors are not linearly related. (See figure 1) Thus, the correlation analysis further confirms that multicollinearity is not a concern among the independent variables.

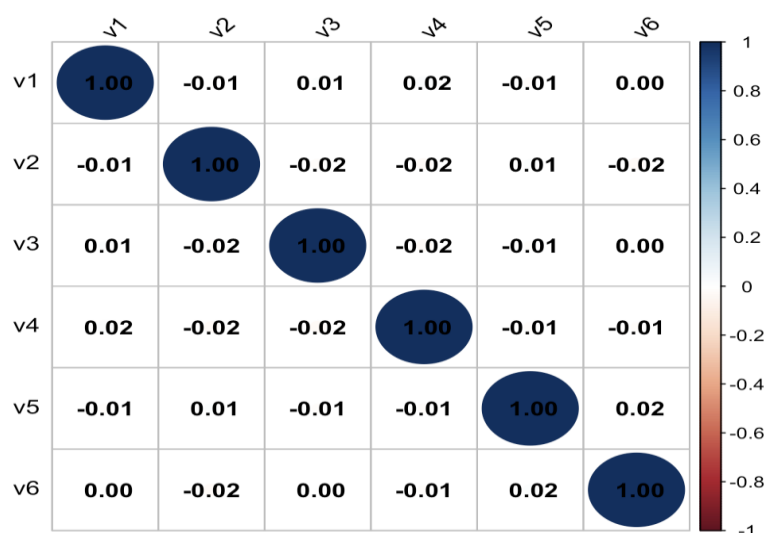


Figure 1 shows a correlation matrix of independent variables from v1-v6

	GVIF	Df	$GVIF^{1/(2*Df)}$
dummyParental_I	1.007610	2	1.001897
dummyAccess_to_Resources	1.009703	2	1.002417
dummyExtracurricular_Activities	1.004177	1	1.002086
dummyMotivation_Level	1.007764	2	1.001935
dummyInternet_Access	1.002873	1	1.001435
dummyFamily_Income	1.007708	2	1.001922
dummyTeacher_Quality	1.006628	2	1.001653
dummySchool_Type	1.002930	1	1.001464
dummyPeer_Influence	1.007398	2	1.001844
dummyLearning_Disabilities	1.002750	1	1.001374
dummyDistance_from_Home	1.004999	2	1.001248
dummyGender	1.002431	1	1.001215
v1	1.002716	1	1.001357
v2	1.004593	1	1.002294
v3	1.002982	1	1.001490
v4	1.005675	1	1.002833
v5	1.001785	1	1.000892
v6	1.006237	1	1.003114

Figure 1.1: VIF Model Test

Step 2: Linearity

To check the linearity assumptions, we followed the steps described in the Student Manual (pp.14-16). Using scatter plots to plot the Exam score (dependent variable) against each independent variable, Hours, Attendance, Sleep Hours, Previous Scores, Tutoring sessions, and Physical activity, showed a clear positive trend. The following plots show a significantly weaker pattern in the plots for Sleep Hours, Tutoring Sessions, and Physical Activity, but none of them suggests a non-linear pattern. Based on the manual's criteria, this means that the linearity assumption is satisfied. (See Figure 2)

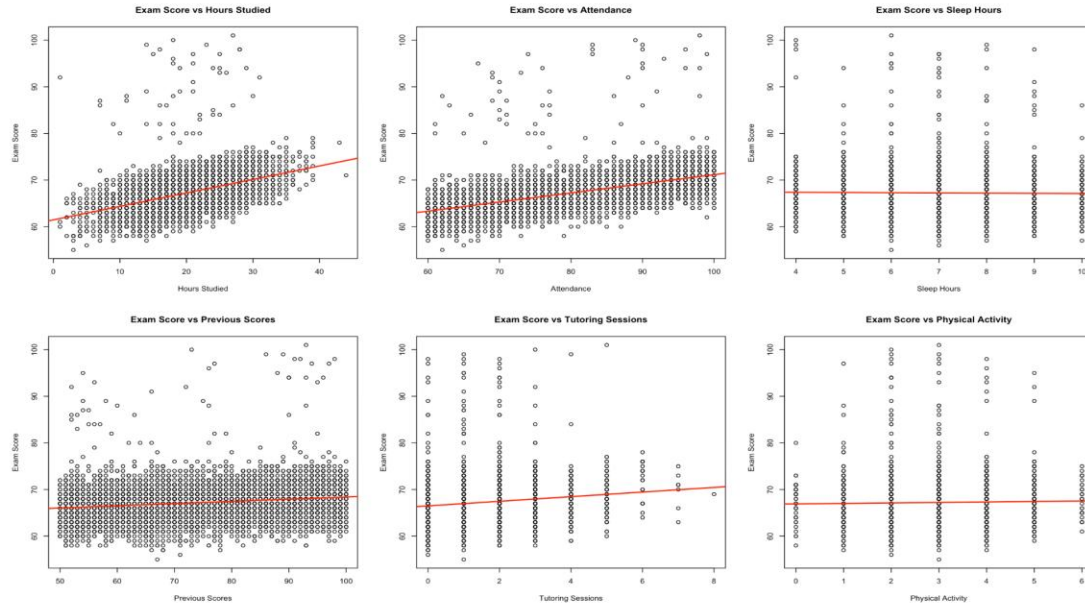


Figure 2: Shows scatter plots between the dependent variable vs each independent variable

Step 3: Fitting the Multiple Regression Model

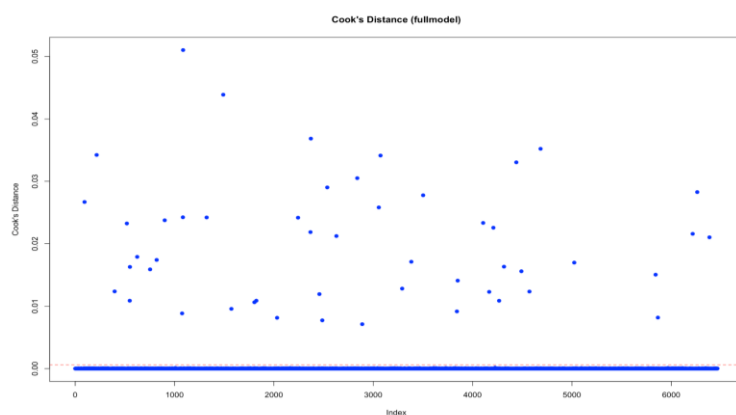
By using the standardised residuals > 3 from Cook's Test, we detected the influential outliers in the model. After removing influential outliers, we re-fitted the model using the cleaned dataset. The new model showed more stable coefficient estimates and improved our diagnostic plots. As per the Student Manual, pages 12-13 suggest that refitting the model after outlier removal ensures no disproportionate influence that can influence the results. The updated

model achieved an R^2 of 0.9772 and an adjusted R^2 of 0.9772, indicating that 97% of the variation of Exam score is explained by the predictors. Thus, the overall fit improved.

Figure 3: Cook's Value plot

The Cook's Distance plot shows that the observations have a very low influence, with no points approaching the cutoff.

- Checking for extreme values



	Index <int>	Residuals <dbl>
95	94	13.670593
218	216	13.230212
405	397	9.605091
530	520	14.184589
559	549	9.611680
561	551	9.081623
638	624	8.971383
771	754	10.799392
837	820	11.993340
920	901	13.852006

Figure 3.1: Shows the R output of the extreme values

We also tested for extreme predictor values visually using a leverage plot. Figure 3.2 shows that none of the students has any high leverage values, suggesting our sample does not contain any extreme predictor values.

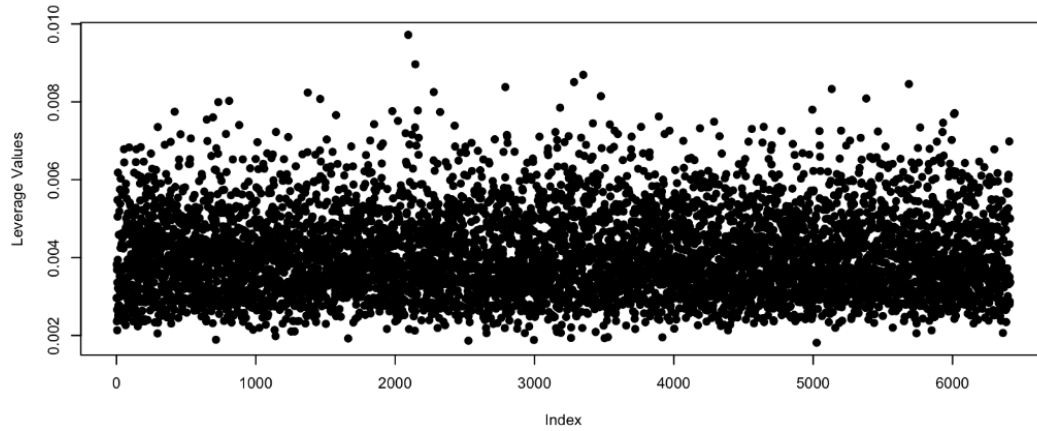


Figure 3.2: Leverage plot for extreme predictor values.

Therefore, the full model is as follows:

Full Model = $\text{lm}(\text{Exam_Score} \sim \text{dummyParental_I} + \text{dummyAccess_to_Resources} +$
 $\text{dummyExtracurricular_Activities} + \text{dummyMotivation_Level} +$
 $\text{dummyInternet_Access} + \text{dummyFamily_Income} + \text{dummyTeacher_Quality} +$
 $\text{dummySchool_Type} + \text{dummyPeer_Influence} + \text{dummyParental_Education_Level} +$
 $\text{dummyLearning_Disabilities} +$
 $\text{dummyDistance_from_Home} + \text{dummyGender} + v1 + v2 + v3 + v4 +$
 $v5 + v6)$

Coefficients:				
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	40.7151998	0.0822430	495.060	< 2e-16 ***
dummyParental_ILow	-1.9835287	0.0184149	-107.713	< 2e-16 ***
dummyParental_IMedium	-1.0089291	0.0147783	-68.271	< 2e-16 ***
dummyAccess_to_ResourcesLow	-2.0011107	0.0183642	-108.968	< 2e-16 ***
dummyAccess_to_ResourcesMedium	-0.9708968	0.0146628	-66.215	< 2e-16 ***
dummyExtracurricular_ActivitiesYes	0.5081617	0.0129215	39.327	< 2e-16 ***
dummyMotivation_LevelLow	-1.0067588	0.0183770	-54.784	< 2e-16 ***
dummyMotivation_LevelMedium	-0.4921695	0.0167349	-29.410	< 2e-16 ***
dummyInternet_AccessYes	0.9996003	0.0239903	41.667	< 2e-16 ***
dummyFamily_IncomeLow	-0.9870540	0.0175717	-56.173	< 2e-16 ***
dummyFamily_IncomeMedium	-0.4799756	0.0176034	-27.266	< 2e-16 ***
dummyTeacher_QualityLow	-1.0019374	0.0230796	-43.412	< 2e-16 ***
dummyTeacher_QualityMedium	-0.5114573	0.0141834	-36.060	< 2e-16 ***
dummySchool_TypePublic	-0.0176602	0.0137770	-1.282	0.200
dummyPeer_InfluenceNeutral	0.4913854	0.0171899	28.586	< 2e-16 ***
dummyPeer_InfluencePositive	1.0063192	0.0171203	58.779	< 2e-16 ***
dummyLearning_DisabilitiesYes	-1.0218311	0.0207038	-49.355	< 2e-16 ***
dummyDistance_from_HomeModerate	0.5019698	0.0231728	21.662	< 2e-16 ***
dummyDistance_from_HomeNear	0.9979812	0.0217437	45.897	< 2e-16 ***
dummyGenderMale	-0.0126021	0.0128324	-0.982	0.326
v1	0.2975218	0.0010587	281.033	< 2e-16 ***
v2	0.2009508	0.0005492	365.925	< 2e-16 ***
v3	0.0199726	0.0043168	4.627	3.79e-06 ***
v4	0.0491056	0.0004413	111.275	< 2e-16 ***
v5	0.5026135	0.0051414	97.758	< 2e-16 ***
v6	0.2229242	0.0061667	36.150	< 2e-16 ***

Figure 5: shows the coefficients of the Full Model and their significance.

Checking the Model Assumptions

- *Endogeneity*

As the student manual does not provide a definitive test for endogeneity, we agreed to visually test for the presence of endogeneity by plotting the residuals vs the covariates. It is stated that we can detect suspicions of endogeneity if there are clustering structures within the residual plot or if the residuals change for values of X. Therefore, we took the extra step of using

the scatter plots and box plots to further our understanding. The red lines represent the trend, which indicates a discernible rising or decreasing trend in the residuals.

Figure 4.1

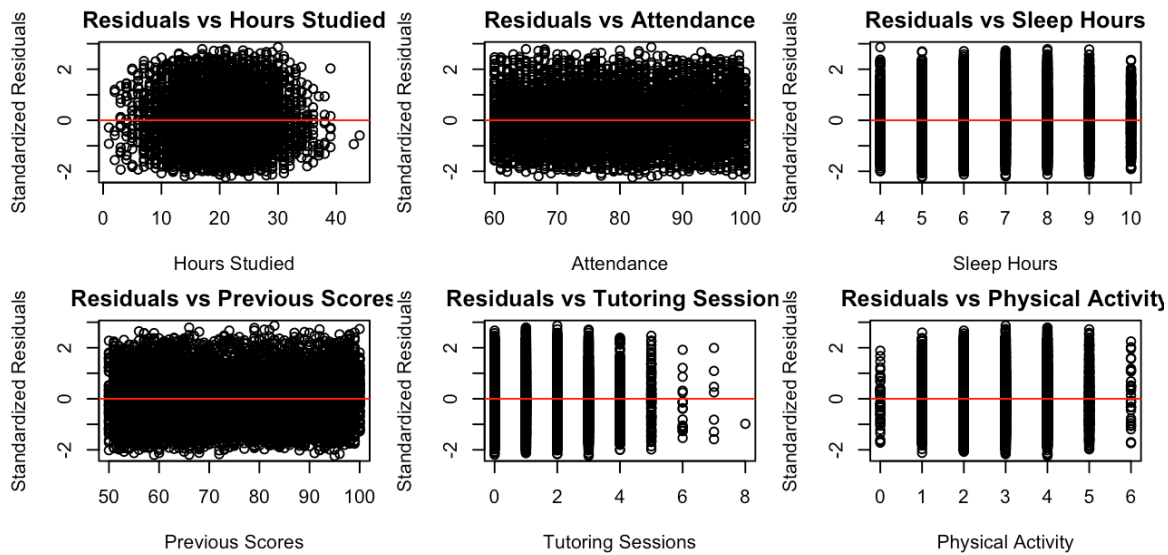


Figure 4.2

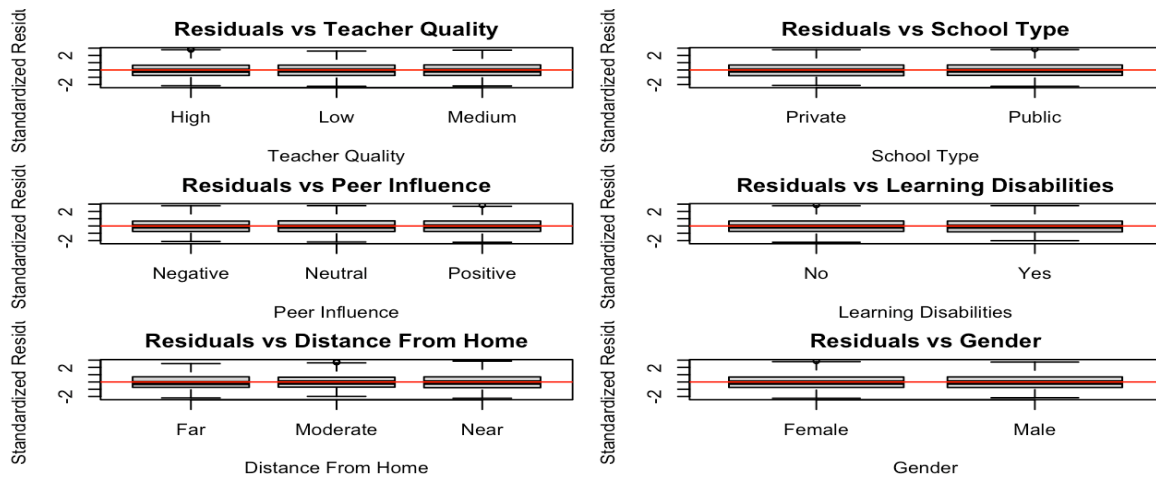
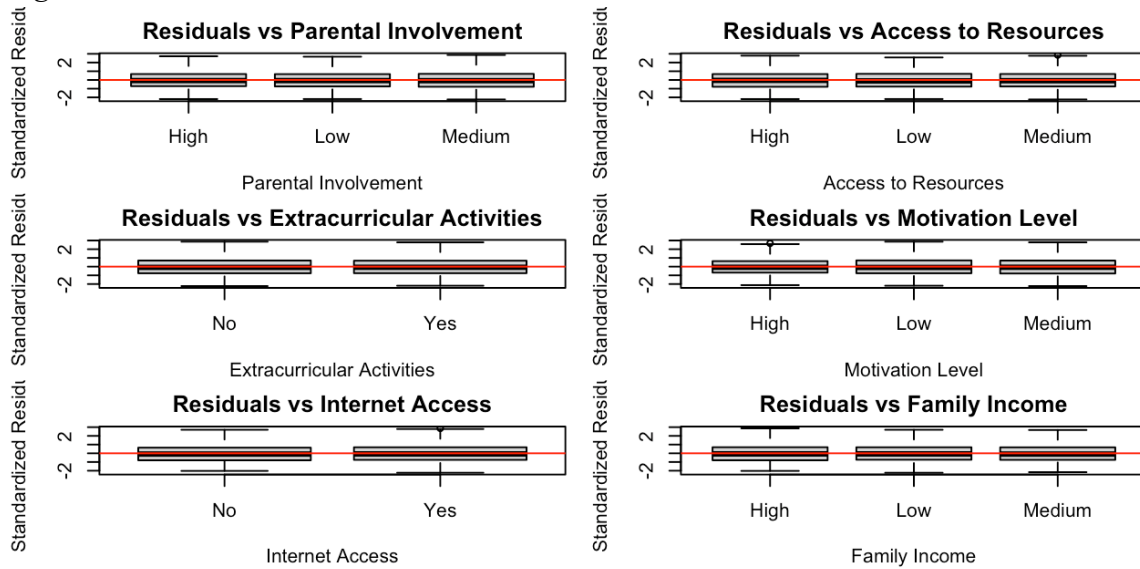


Figure 4.3



Similar comparisons between the residuals and our categorical variables are shown in Figures 4.2 and 4.3. Visually, there are no discernible differences between our residuals and the categorical variables.

Checking assumptions related to the residuals

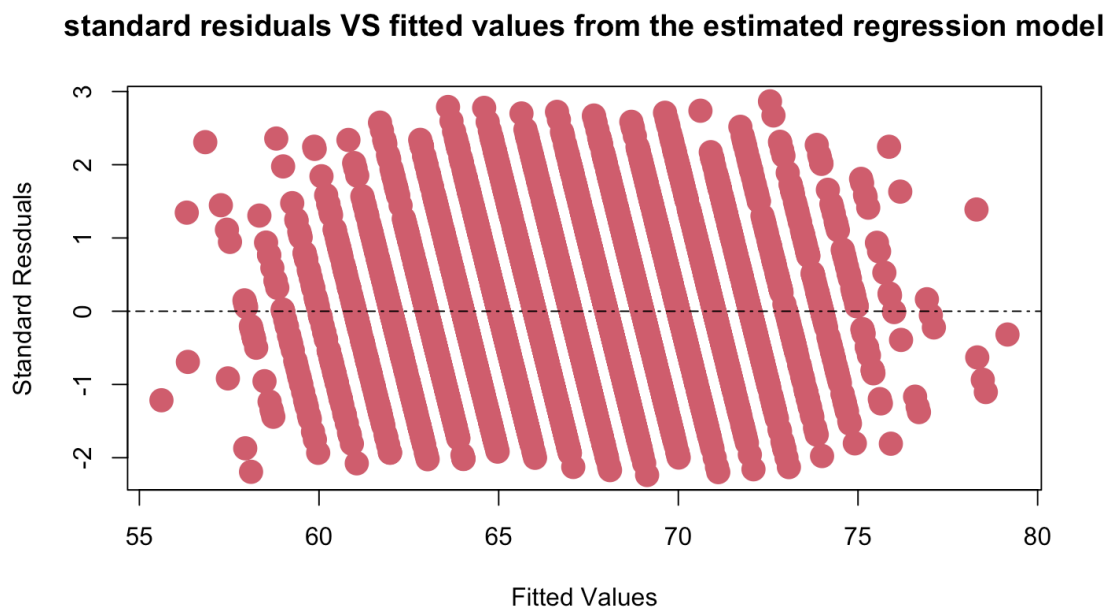


Figure 5: Plot of the residuals vs the full model

The plot shown in Figure 5 suggests that the residuals are scattered around the zero mean. However, the striped patterns and non-constant spreads suggest potential heteroscedasticity. However, this is likely due to the structure of our dependent variable (0 to 100) and is not a clear indicator of heteroscedasticity. Visually, there are no changes in the spread of the residuals around the fitted values, from which we conclude that the assumption of heteroscedasticity has not been violated.

- *Normality of Errors*

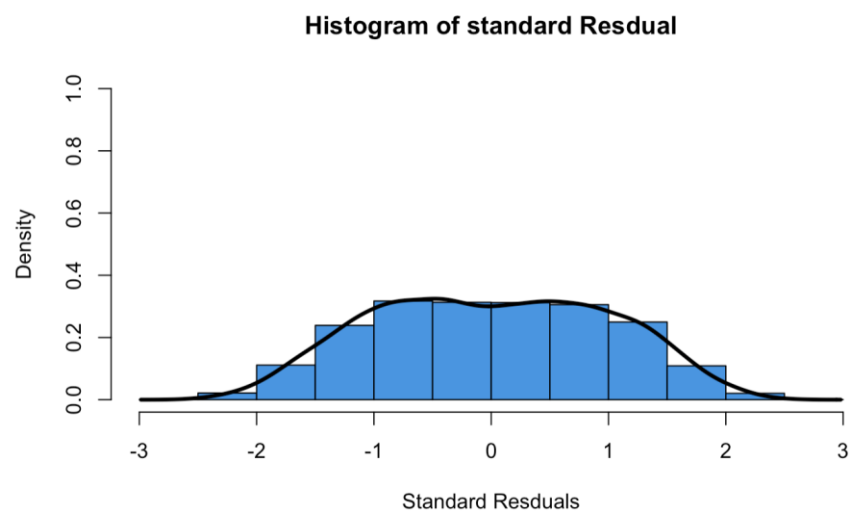


Figure 6.1: Histogram of standard residuals

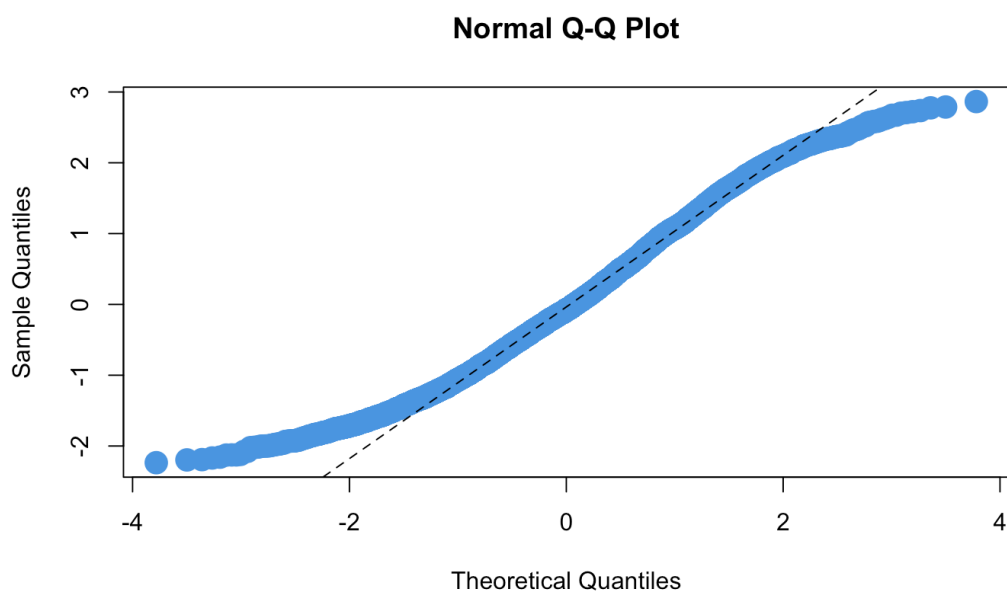


Figure 6.2: Normal QQ-plot

Figures 6.1 and 6.2 show the normal distribution of errors. The histogram displayed a roughly symmetric and bell-shaped distribution around zero, indicating no skewness or kurtosis. The Q-Q plot further supports the evidence of normal distribution in the central region and small deviations from the tail. Overall, Figures 6.1 and 6.2 provide strong evidence that the residuals satisfy the normality assumptions needed for valid inference in linear regression (Student Manual, page 110).

- Checking for the Presence of Autocorrelation

Given our sample of students was randomly selected, the observations are likely independently and identically distributed (i.i.d.), and we did not expect to find any evidence of autocorrelation. As per the methodology of the Student Manual, the Durbin-Watson test was conducted to examine the independence of the residuals. The test produces a statistic of $DW = 2.00085$ with a corresponding p-value = 0.6334. A DW near 2 indicates the absence of both positive and negative autocorrelation. Given the P value, we fail to reject the null hypothesis of no autocorrelation. Overall, we conclude that the independence assumption was not violated.

Model Selection

- Methods of model selection

1. Backward Elimination

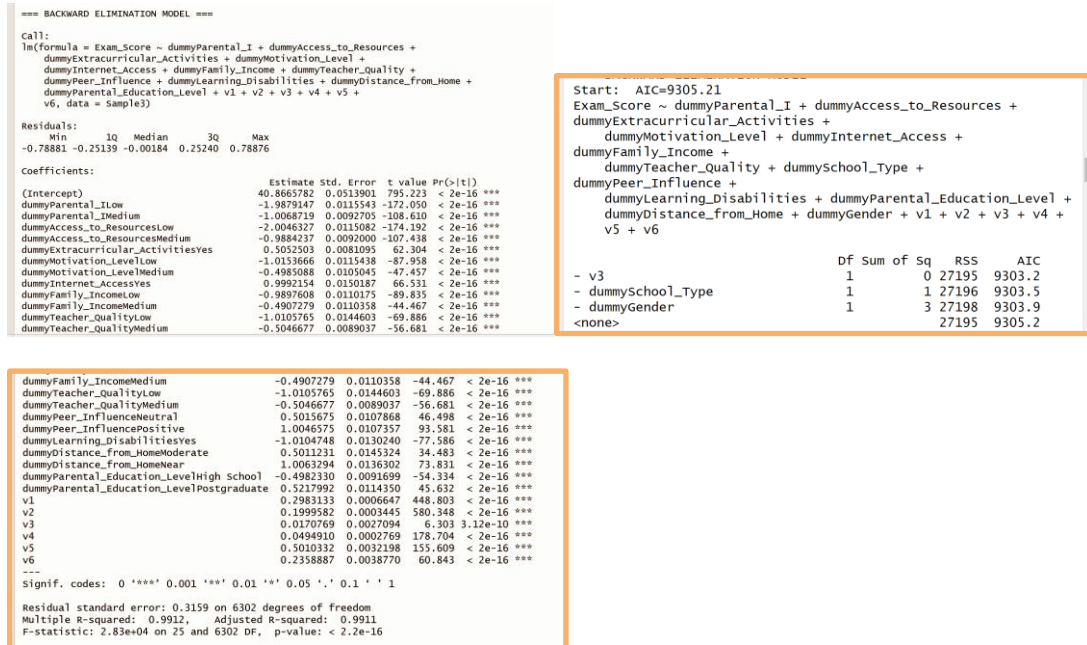


Figure: Summary of **Backwards Elimination Method**

We can see that in backward Elimination, our full model contains

Full Model = $\text{lm}(\text{Exam_Score} \sim \text{dummyParental_I} + \text{dummyAccess_to_Resources} +$
 $\text{dummyExtracurricular_Activities} + \text{dummyMotivation_Level} +$
 $\text{dummyInternet_Access} + \text{dummyFamily_Income} + \text{dummyTeacher_Quality} +$
 $\text{dummyPeer_Influence} + \text{dummyParental_Education_Level} + \text{dummyLearning_Disabilities} +$
 $\text{dummyDistance_from_Home} + v1 + v2 + v3 + v4 + v5 + v6)$

In the figure, we notice that **Backwards Elimination** chooses to eliminate 3 variables that have the most insignificant R^2 with the highest p-value, which are (dummySchool_Type,

dummyGender, and V3 Sleep_hours). We can see that Sleep_hours was included in the full model. Although its statistical effect appears small, it is theoretically meaningful. Excluding it solely based on its weak contribution could reduce the model's interpretability. In addition, removing this variable may slightly increase the variance inflation of the remaining predictors.

2. Forward Selection

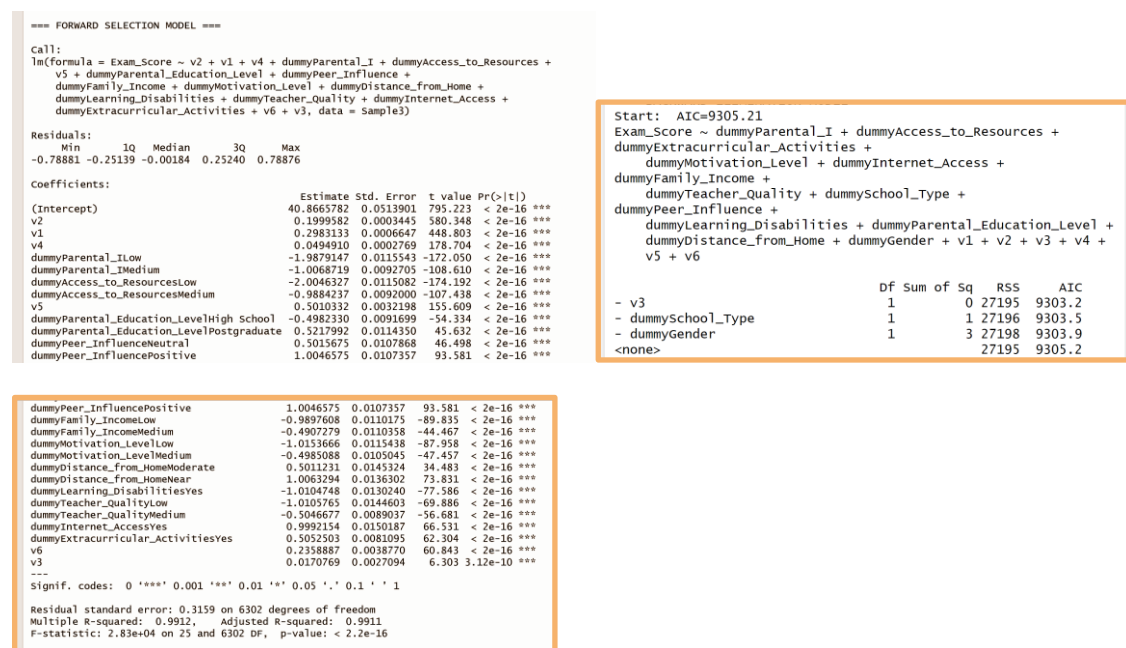


Figure: Summary of the Forward Selection Method

Here we can see that the **Forward Selection** we can observe that forward selection chooses the variables that produce the highest R^2 , but in this method we notice that it chooses to eliminate 3 variables which they are (dummySchool_Type, dummyGender, and V3 Sleep_hours), we can see that Sleep_hours was included in the full model because eliminating this variable might slightly increase variance inflation for other variables and could weaken the model's interpretability.

3. Stepwise Selection

```

=== STEPWISE MODEL ===

Call:
lm(formula = Exam_Score ~ dummyParental_I + dummyAccess_to_Resources +
  dummyExtracurricular_Activities + dummyMotivation_Level +
  dummyInternet_Access + dummyFamily_Income + dummyTeacher_Quality +
  dummyPeer_Influence + dummyLearning_Disabilities + dummyDistance_from_Home +
  dummyParental_Education_Level + v1 + v2 + v3 + v4 + v5 +
  v6, data = Sample3)

Residuals:
    Min       1Q   Median       3Q      Max
-0.78881 -0.25139 -0.00184  0.25240  0.78876

```

Figure: Summary of Stepwise Selection Method

We notice that in the 3rd method of **Stepwise Selection**, we have the same variables eliminated as well as the same variables added, like in the previous two methods mentioned above, although they have different approaches, but they give the same FINAL full model result.

IDENTIFY BEST MODEL

- Comparing

Model <chr>	AIC <dbl>	BIC <dbl>	Adjusted_R2 <dbl>
Full	3403.02	3598.85	0.9911
Backward	3400.30	3582.62	0.9911
Forward	3400.30	3582.62	0.9911
Stepwise	3400.30	3582.62	0.9911

4 rows

Figure: Summary of AIC, BIC, and R^2

We compare each model method using AIC, BIC, and R^2 to determine the best-fitting model method, because different model selection methods do not always produce the same results.

• *The best-fitted model*

```

===== BEST MODELS =====
Lowest AIC Model: Backward
Lowest BIC Model: Backward
Highest Adjusted R2 Model: Backward

===== FINAL SELECTED MODEL (Stepwise) =====

Call:
lm(formula = Exam_Score ~ dummyParental_I +
  dummyAccess_to_Resources +
    dummyExtracurricular_Activities + dummyMotivation_Level +
    dummyInternet_Access + dummyFamily_Income +
  dummyTeacher_Quality +
    dummyPeer_Influence + dummyLearning_Disabilities +
  dummyDistance_from_Home +
    dummyParental_Education_Level + v1 + v2 + v3 + v4 + v5 +
    v6, data = Sample3)

```

Figure: Summary of the best-fitted model

Although backward elimination produced the lowest AIC, BIC, and highest R^2 , all three selection methods (forward, backward, and stepwise) produced identical results. Therefore, since all models are equivalent, we selected stepwise selection as the final model, as it is the standard approach that combines both forward and backward selection, making it the most methodologically justified choice.

Therefore, the full model is as follows:

Full Model = $\text{lm}(\text{Exam_Score} \sim \text{dummyParental_I} + \text{dummyAccess_to_Resources} +$
 $\text{dummyExtracurricular_Activities} + \text{dummyMotivation_Level} +$
 $\text{dummyInternet_Access} + \text{dummyFamily_Income} + \text{dummyTeacher_Quality} +$
 $\text{dummyPeer_Influence} + \text{dummyParental_Education_Level} + \text{dummyLearning_Disabilities} +$
 $\text{dummyDistance_from_Home} + v1 + v2 + v3 + v4 + v5 + v6)$

Final Model Reporting

===== FINAL SELECTED MODEL (Stepwise) =====

Call:

```
lm(formula = Exam_Score ~ dummyParental_I + dummyAccess_to_Resources +
    dummyExtracurricular_Activities + dummyMotivation_Level +
    dummyInternet_Access + dummyFamily_Income + dummyTeacher_Quality +
    dummyPeer_Influence + dummyLearning_Disabilities + dummyDistance_from_Home +
    dummyParental_Education_Level + v1 + v2 + v3 + v4 + v5 +
    v6, data = sample3)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.78881	-0.25139	-0.00184	0.25240	0.78876

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	40.8665782	0.0513901	795.223	< 2e-16	***
dummyParental_ILow	-1.9879147	0.0115543	-172.050	< 2e-16	***
dummyParental_IMedium	-1.0068719	0.0092705	-108.610	< 2e-16	***
dummyAccess_to_ResourcesLow	-2.0046327	0.0115082	-174.192	< 2e-16	***
dummyAccess_to_ResourcesMedium	-0.9884237	0.0092000	-107.438	< 2e-16	***
dummyExtracurricular_ActivitiesYes	0.5052503	0.0081095	62.304	< 2e-16	***
dummyMotivation_LevelLow	-1.0153666	0.0115438	-87.958	< 2e-16	***
dummyMotivation_LevelMedium	-0.4985088	0.0105045	-47.457	< 2e-16	***
dummyInternet_AccessYes	0.9992154	0.0150187	66.531	< 2e-16	***
dummyFamily_IncomeLow	-0.9897608	0.0110175	-89.835	< 2e-16	***
dummyFamily_IncomeMedium	-0.4907279	0.0110358	-44.467	< 2e-16	***
dummyTeacher_QualityLow	-1.0105765	0.0144603	-69.886	< 2e-16	***
dummyTeacher_QualityMedium	-0.5046677	0.0089037	-56.681	< 2e-16	***
dummyPeer_InfluenceNeutral	0.5015675	0.0107868	46.498	< 2e-16	***
dummyPeer_InfluencePositive	1.0046575	0.0107357	93.581	< 2e-16	***
dummyLearning_DisabilitiesYes	-1.0104748	0.0130240	-77.586	< 2e-16	***
dummyDistance_from_HomeModerate	0.5011231	0.0145324	34.483	< 2e-16	***
dummyDistance_from_HomeNear	1.0063294	0.0136302	73.831	< 2e-16	***
dummyParental_Education_LevelHigh school	-0.4982330	0.0091699	-54.334	< 2e-16	***
dummyParental_Education_LevelPostgraduate	0.5217992	0.0114350	45.632	< 2e-16	***
v1	0.2983133	0.0006647	448.803	< 2e-16	***
v2	0.1999582	0.0003445	580.348	< 2e-16	***
v3	0.0170769	0.0027094	6.303	3.12e-10	***
v4	0.0494910	0.0002769	178.704	< 2e-16	***
v5	0.5010332	0.0032198	155.609	< 2e-16	***
v6	0.2358887	0.0038770	60.843	< 2e-16	***

signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3159 on 6302 degrees of freedom

Multiple R-squared: 0.9912, Adjusted R-squared: 0.9911

F-statistic: 2.83e+04 on 25 and 6302 DF, p-value: < 2.2e-16

Figure: Summary of the Best and Final Model

Coefficient Interpretation

Interpretation of Each Coefficient

Parental Involvement

- Low: Students with low parental involvement score 1.9879 points lower than those with high involvement.
- Medium: Students with a medium involvement score 1.0069 points lower than those with high involvement.
- Higher parental involvement is strongly linked to higher exam scores.

Access to Resources

- Low Access: Students score 2.0046 points lower than high-access students.
- Medium Access: Students score 0.9884 points lower than high-access students.
- Access to learning resources has a clear positive impact on exam performance.

Extracurricular Activities

- Yes: Students who participate in extracurricular activities score 0.5053 points higher than those who do not.
- Extracurricular involvement modestly boosts academic outcomes.

Motivation Level

- Low Motivation: Students score 1.0154 points lower compared to highly motivated students.
- Medium Motivation: Students score 0.4986 points lower than highly motivated students.
- Motivation is one of the strongest individual predictors of exam performance.

Internet Access

- Yes: Students with home internet access score 0.9992 points higher than those without it.
- Internet access strongly supports academic achievement.

Family Income

- Low Income: Students score 0.9898 points lower than high-income students.
- Medium Income: Students score 0.4907 points lower than high-income students.
- Higher socioeconomic status positively impacts exam performance.

Teacher Quality

- Low Teacher Quality: Students score 1.0105 points lower than those taught by high-quality teachers.
- Medium Teacher Quality: Students score 0.5047 points lower than those taught by high-quality teachers.
- Better teacher quality consistently improves outcomes.

Peer Influence

- Neutral Peer Influence: Students score 0.5016 points higher than those with negative peer influence.
- Positive Peer Influence: Students score 1.0047 points higher than those with negative peer influence.
- Positive peer environments substantially support learning.

Learning Disabilities

- Yes: Students with learning disabilities score 1.0105 points lower than students without learning disabilities.
- This reflects academic challenges associated with learning disabilities.

Distance From Home

- Moderate Distance: Students score 0.5011 points higher than those who live far from school.
- Near Distance: Students score 1.0063 points higher than those who live far.
- Unexpectedly, students living far away perform best—possibly due to unobserved factors (e.g., school quality variations).

Parental Education Level

- High School: Students score 0.4982 points lower compared to students whose parents' highest education is an undergraduate degree (baseline).
- Postgraduate: Students score 0.5214 points higher than those with undergraduate-educated parents.
- Higher parental education corresponds to higher exam scores.

Statistical Analysis of the Model

Hypothesis Tests

For every coefficient in the model, the hypotheses follow this structure:

$H_0: \beta_i = 0$ (the predictor does not affect Exam Score)

$H_1: \beta_i \neq 0$ (the predictor has a significant effect)

- Parental Involvement

```
call:
lm(formula = Exam_Score ~ dummyAccess_to_Resources + dummyExtracurricular_Activities +
  dummyMotivation_Level + dummyInternet_Access + dummyFamily_Income +
  dummyTeacher_Quality + dummyPeer_Influence + dummyLearning_Disabilities +
  dummyDistance_from_Home + dummyParental_Education_Level +
  v1 + v2 + v3 + v4 + v5 + v6, data = Sample3)

Residuals:
    Min       1Q   Median       3Q      Max
-1.90923 -0.51422 -0.00319  0.55604  1.66873

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    40.0397434   0.1227434  326.207 <2e-16 ***
dummyAccess_to_ResourcesLow -1.9381250   0.0276583  -70.074 <2e-16 ***
dummyAccess_to_ResourcesMedium -0.9938846   0.0221229  -44.926 <2e-16 ***
dummyExtracurricular_ActivitiesYes  0.4699495   0.0194934   24.108 <2e-16 ***
dummyMotivation_LevelLow -0.9715124   0.0277528  -35.006 <2e-16 ***
dummyMotivation_LevelMedium -0.4760600   0.0252580  -18.848 <2e-16 ***
dummyInternet_AccessYes  1.0372608   0.0361110   28.724 <2e-16 ***
dummyFamily_IncomeLow -1.0011585   0.0264903  -37.793 <2e-16 ***
dummyFamily_IncomeMedium -0.4871473   0.0265380  -18.357 <2e-16 ***
dummyTeacher_QualityLow -1.0619252   0.0347657  -30.545 <2e-16 ***
dummyTeacher_QualityMedium -0.5087968   0.0214108  -23.764 <2e-16 ***
dummyPeer_InfluenceNeutral  0.5264894   0.0259333   20.302 <2e-16 ***
dummyPeer_InfluencePositive  1.0464844   0.0258095   40.546 <2e-16 ***
dummyLearning_DisabilitiesYes -0.9912401   0.0313165  -31.652 <2e-16 ***
dummyDistance_from_HomeModerate  0.5257784   0.0349444   15.046 <2e-16 ***
dummyDistance_from_HomeNear  1.0295432   0.0327752   31.412 <2e-16 ***
dummyParental_Education_LevelHigh School -0.5038623   0.0220473  -22.854 <2e-16 ***
dummyParental_Education_LevelPostgraduate  0.5028479   0.0274955   18.288 <2e-16 ***
v1              0.2965766   0.0015982  185.568 <2e-16 ***
v2              0.1995010   0.0008285  240.807 <2e-16 ***
v3              0.0146882   0.0065154    2.254  0.0242 *
v4              0.0485455   0.0006658   72.912 <2e-16 ***
v5              0.4986016   0.0077424   64.399 <2e-16 ***
v6              0.2334459   0.0093229   25.040 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7595 on 6304 degrees of freedom
Multiple R-squared:  0.9489, Adjusted R-squared:  0.9487
F-statistic: 5093 on 23 and 6304 DF, p-value: < 2.2e-16
```


- Access to Resources

```
Call:
lm(formula = Exam_Score ~ dummyParental_I + dummyExtracurricular_Activities +
  dummyMotivation_Level + dummyInternet_Access + dummyFamily_Income +
  dummyTeacher_Quality + dummyPeer_Influence + dummyLearning_Disabilities +
  dummyDistance_from_Home + dummyParental_Education_Level +
  v1 + v2 + v3 + v4 + v5 + v6, data = Sample3)

Residuals:
    Min       1Q   Median       3Q      Max
-1.89122 -0.50337  0.00747  0.56113  1.63400

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  40.0605157  0.1241089  322.785 <2e-16 ***
dummyParental_ILow -1.9261005  0.0280391  -68.693 <2e-16 ***
dummyParental_IMedium -0.9839331  0.0225091  -43.713 <2e-16 ***
dummyExtracurricular_ActivitiesYes  0.4851413  0.0196914   24.637 <2e-16 ***
dummyMotivation_LevelLow -1.0270303  0.0280344  -36.635 <2e-16 ***
dummyMotivation_LevelMedium -0.4785904  0.0255105  -18.761 <2e-16 ***
dummyInternet_AccessYes  0.9541629  0.0364709   26.162 <2e-16 ***
dummyFamily_IncomeLow -0.9886594  0.0267571  -36.949 <2e-16 ***
dummyFamily_IncomeMedium -0.5152604  0.0268007  -19.226 <2e-16 ***
dummyTeacher_QualityLow -0.9930630  0.0351187  -28.277 <2e-16 ***
dummyTeacher_QualityMedium -0.4881998  0.0216210  -22.580 <2e-16 ***
dummyPeer_InfluenceNeutral  0.4806648  0.0261960   18.349 <2e-16 ***
dummyPeer_InfluencePositive  1.0004888  0.0260726   38.373 <2e-16 ***
dummyLearning_DisabilitiesYes -1.0215313  0.0316311  -32.295 <2e-16 ***
dummyDistance_from_HomeModerate  0.5180426  0.0352843   14.682 <2e-16 ***
dummyDistance_from_HomeNear  1.0163289  0.0330901   30.714 <2e-16 ***
dummyParental_Education_LevelHigh School -0.4987604  0.0222643  -22.402 <2e-16 ***
dummyParental_Education_LevelPostgraduate  0.5130631  0.0277716   18.474 <2e-16 ***
v1  0.2979108  0.0016141  184.563 <2e-16 ***
v2  0.1991412  0.0008367  237.998 <2e-16 ***
v3  0.0105069  0.0065793    1.597  0.11
v4  0.0506900  0.0006724   75.390 <2e-16 ***
v5  0.4930188  0.0078193   63.052 <2e-16 ***
v6  0.2267981  0.0094089   24.105 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7671 on 6304 degrees of freedom
Multiple R-squared:  0.9479,    Adjusted R-squared:  0.9477
F-statistic: 4987 on 23 and 6304 DF,  p-value: < 2.2e-16
```

- Extracurricular Activities

```
Call:
lm(formula = Exam_Score ~ dummyParental_I + dummyAccess_to_Resources +
  dummyMotivation_Level + dummyInternet_Access + dummyFamily_Income +
  dummyTeacher_Quality + dummyPeer_Influence + dummyLearning_Disabilities +
  dummyDistance_from_Home + dummyParental_Education_Level +
  v1 + v2 + v3 + v4 + v5 + v6, data = Sample3)

Residuals:
    Min       1Q   Median       3Q      Max
-1.08563 -0.27595  0.00584  0.29633  0.96329

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  41.1656500  0.0650364  632.963 < 2e-16 ***
dummyParental_ILow -1.9712601  0.0146827  -134.257 < 2e-16 ***
dummyParental_IMedium -0.9921374  0.0117799  -84.223 < 2e-16 ***
dummyAccess_to_ResourcesLow -1.9931265  0.0146262  -136.271 < 2e-16 ***
dummyAccess_to_ResourcesMedium -0.9905456  0.0116940  -84.705 < 2e-16 ***
dummyMotivation_LevelLow -1.0253315  0.0146719  -69.884 < 2e-16 ***
dummyMotivation_LevelMedium -0.5028221  0.0133519  -37.659 < 2e-16 ***
dummyInternet_AccessYes  0.9937189  0.0190899   52.055 < 2e-16 ***
dummyFamily_IncomeLow -0.9832534  0.0140038  -70.214 < 2e-16 ***
dummyFamily_IncomeMedium -0.4816514  0.0140264  -34.339 < 2e-16 ***
dummyTeacher_QualityLow -1.0283341  0.0183769  -55.958 < 2e-16 ***
dummyTeacher_QualityMedium -0.5072852  0.0113173  -44.824 < 2e-16 ***
dummyPeer_InfluenceNeutral  0.4871077  0.0137080   35.535 < 2e-16 ***
dummyPeer_InfluencePositive  1.0063547  0.0136461   73.747 < 2e-16 ***
dummyLearning_DisabilitiesYes -1.0185835  0.0165540  -61.531 < 2e-16 ***
dummyDistance_from_HomeModerate  0.5178730  0.0184689   28.040 < 2e-16 ***
dummyDistance_from_HomeNear  1.0081806  0.0173253   58.191 < 2e-16 ***
dummyParental_Education_LevelHigh School -0.5067549  0.0116545  -43.481 < 2e-16 ***
dummyParental_Education_LevelPostgraduate  0.5125724  0.0145339   35.267 < 2e-16 ***
v1  0.2979639  0.0008449  352.681 < 2e-16 ***
v2  0.1999378  0.0004380  456.525 < 2e-16 ***
v3  0.0177772  0.0034439    5.162 2.52e-07 ***
v4  0.0495734  0.0003520  140.826 < 2e-16 ***
v5  0.5021736  0.0040926  122.701 < 2e-16 ***
v6  0.2354904  0.0049281   47.786 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4015 on 6303 degrees of freedom
Multiple R-squared:  0.9857,    Adjusted R-squared:  0.9857
F-statistic: 1.815e+04 on 24 and 6303 DF,  p-value: < 2.2e-16
```

- Motivation Level

```
Call:
lm(formula = Exam_Score ~ dummyParental_I + dummyAccess_to_Resources +
  dummyExtracurricular_Activities + dummyInternet_Access +
  dummyFamily_Income + dummyTeacher_Quality + dummyPeer_Influence +
  dummyLearning_Disabilities + dummyDistance_from_Home + dummyParental_Education_Level +
  v1 + v2 + v3 + v4 + v5 + v6, data = Sample3)

Residuals:
    Min       1Q   Median       3Q      Max
-1.24920 -0.33004 -0.00867  0.34186  1.33975

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  40.3044995  0.0762343  528.693 < 2e-16 ***
dummyParental_ILow -1.9649376  0.0173434 -113.296 < 2e-16 ***
dummyParental_IMedium -0.9980928  0.0139177 -71.714 < 2e-16 ***
dummyAccess_to_ResourcesLow -2.0148241  0.0172750 -116.633 < 2e-16 ***
dummyAccess_to_ResourcesMedium -0.9852741  0.0138120 -71.335 < 2e-16 ***
dummyExtracurricular_ActivitiesYes 0.5154775  0.0121745  42.341 < 2e-16 ***
dummyInternet_AccessYes 1.0246132  0.0225447  45.448 < 2e-16 ***
dummyFamily_IncomeLow -0.9939977  0.0165417 -60.091 < 2e-16 ***
dummyFamily_IncomeMedium -0.4793756  0.0165682 -28.933 < 2e-16 ***
dummyTeacher_QualityLow -0.9799621  0.0216932 -45.174 < 2e-16 ***
dummyTeacher_QualityMedium -0.5116617  0.0133666 -38.279 < 2e-16 ***
dummyPeer_InfluenceNeutral 0.5033001  0.0161909  31.085 < 2e-16 ***
dummyPeer_InfluencePositive 1.0057999  0.0161181  62.402 < 2e-16 ***
dummyLearning_DisabilitiesYes -1.0202416  0.0195501 -52.186 < 2e-16 ***
dummyDistance_from_HomeModerate 0.4803457  0.0218161  22.018 < 2e-16 ***
dummyDistance_from_HomeNear 0.9937392  0.0204634  48.562 < 2e-16 ***
dummyParental_Education_LevelHigh School -0.5024790  0.0137662 -36.501 < 2e-16 ***
dummyParental_Education_LevelPostgraduate 0.5081220  0.0171637  29.604 < 2e-16 ***
v1 0.2977210  0.0009978  298.380 < 2e-16 ***
v2 0.1999390  0.0005173  386.513 < 2e-16 ***
v3 0.0172925  0.0040680  4.251 2.16e-05 ***
v4 0.0496462  0.0004157  119.433 < 2e-16 ***
v5 0.5028160  0.0048339  104.019 < 2e-16 ***
v6 0.2318763  0.0058205  39.838 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4742 on 6304 degrees of freedom
Multiple R-squared:  0.9801,    Adjusted R-squared:  0.98
F-statistic: 1.349e+04 on 23 and 6304 DF, p-value: < 2.2e-16
```

- Internet Access

```
Call:
lm(formula = Exam_Score ~ dummyParental_I + dummyAccess_to_Resources +
  dummyExtracurricular_Activities + dummyMotivation_Level +
  dummyFamily_Income + dummyTeacher_Quality + dummyPeer_Influence +
  dummyLearning_Disabilities + dummyDistance_from_Home + dummyParental_Education_Level +
  v1 + v2 + v3 + v4 + v5 + v6, data = Sample3)

Residuals:
    Min       1Q   Median       3Q      Max
-1.65637 -0.23718  0.02769  0.30642  0.86979

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  41.8161206  0.0644087  649.231 < 2e-16 ***
dummyParental_ILow -1.9997975  0.0150725 -132.679 < 2e-16 ***
dummyParental_IMedium -1.0089281  0.0120947 -83.419 < 2e-16 ***
dummyAccess_to_ResourcesLow -1.9913737  0.0150119 -132.653 < 2e-16 ***
dummyAccess_to_ResourcesMedium -0.9834918  0.0120024 -81.941 < 2e-16 ***
dummyExtracurricular_ActivitiesYes 0.5020810  0.0105799  47.456 < 2e-16 ***
dummyMotivation_LevelLow -1.0290526  0.0150582 -68.338 < 2e-16 ***
dummyMotivation_LevelMedium -0.5016444  0.0137045 -36.604 < 2e-16 ***
dummyFamily_IncomeLow -0.9871301  0.0143739 -68.675 < 2e-16 ***
dummyFamily_IncomeMedium -0.4810087  0.0143966 -33.411 < 2e-16 ***
dummyTeacher_QualityLow -1.0102566  0.0188656 -53.550 < 2e-16 ***
dummyTeacher_QualityMedium -0.5062114  0.0116162 -43.578 < 2e-16 ***
dummyPeer_InfluenceNeutral 0.4952906  0.0140725  35.196 < 2e-16 ***
dummyPeer_InfluencePositive 0.9953532  0.0140051  71.071 < 2e-16 ***
dummyLearning_DisabilitiesYes -1.0058839  0.0169915 -59.199 < 2e-16 ***
dummyDistance_from_HomeModerate 0.4992853  0.0189596  26.334 < 2e-16 ***
dummyDistance_from_HomeNear 1.0016190  0.0177824  56.327 < 2e-16 ***
dummyParental_Education_LevelHigh School -0.4853838  0.0119608 -40.581 < 2e-16 ***
dummyParental_Education_LevelPostgraduate 0.5371950  0.0149157  36.015 < 2e-16 ***
v1 0.2986682  0.0008672  344.422 < 2e-16 ***
v2 0.1995065  0.0004494  443.911 < 2e-16 ***
v3 0.0191280  0.0035346  5.412 6.48e-08 ***
v4 0.0495398  0.0003613  137.110 < 2e-16 ***
v5 0.4993812  0.0042006  118.883 < 2e-16 ***
v6 0.2320735  0.0050576  45.886 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4121 on 6303 degrees of freedom
Multiple R-squared:  0.985,    Adjusted R-squared:  0.9849
F-statistic: 1.721e+04 on 24 and 6303 DF, p-value: < 2.2e-16
```

- Family Income

```
Call:
lm(formula = Exam_Score ~ dummyParental_I + dummyAccess_to_Resources +
  dummyExtracurricular_Activities + dummyMotivation_Level +
  dummyInternet_Access + dummyTeacher_Quality + dummyPeer_Influence +
  dummyLearning_Disabilities + dummyDistance_from_Home + dummyParental_Education_Level +
  v1 + v2 + v3 + v4 + v5 + v6, data = Sample3)

Residuals:
    Min       1Q   Median       3Q      Max
-1.18431 -0.34840 -0.01553  0.34564  1.37304

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    40.3510312    0.0782301   515.799 < 2e-16 ***
dummyParental_ILow -1.9988737    0.0177446  -112.647 < 2e-16 ***
dummyParental_IMedium -0.9981457    0.0142376   -70.106 < 2e-16 ***
dummyAccess_to_ResourcesLow -2.0013926    0.0176731  -113.245 < 2e-16 ***
dummyAccess_to_ResourcesMedium -0.9770291    0.0141265   -69.163 < 2e-16 ***
dummyExtracurricular_ActivitiesYes  0.5000125    0.0124542   40.148 < 2e-16 ***
dummyMotivation_LevelLow -1.0255221    0.0177261  -57.854 < 2e-16 ***
dummyMotivation_LevelMedium -0.5067742    0.0161317  -31.415 < 2e-16 ***
dummyInternet_AccessYes  0.9993008    0.0230644   43.327 < 2e-16 ***
dummyTeacher_QualityLow -1.0123820    0.0222055  -45.592 < 2e-16 ***
dummyTeacher_QualityMedium -0.4935339    0.0136738  -36.093 < 2e-16 ***
dummyPeer_InfluenceNeutral  0.5136984    0.0165644   31.012 < 2e-16 ***
dummyPeer_InfluencePositive  1.0224391    0.0164861   62.018 < 2e-16 ***
dummyLearning_DisabilitiesYes -0.9928685    0.0200011  -49.641 < 2e-16 ***
dummyDistance_from_HomeModerate  0.5110272    0.0223141   22.901 < 2e-16 ***
dummyDistance_from_HomeNear  1.0226106    0.0209319   48.854 < 2e-16 ***
dummyParental_Education_LevelHigh School -0.4967186    0.0140829  -35.271 < 2e-16 ***
dummyParental_Education_LevelPostgraduate  0.5257685    0.0175628   29.936 < 2e-16 ***
v1              0.2984894    0.0010208  292.411 < 2e-16 ***
v2              0.1995148    0.0005291  377.062 < 2e-16 ***
v3              0.0135374    0.0041608    3.254 0.00115 **
v4              0.0490674    0.0004253  115.380 < 2e-16 ***
v5              0.5019767    0.0049453  101.507 < 2e-16 ***
v6              0.2262767    0.0059473   38.047 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4851 on 6304 degrees of freedom
Multiple R-squared:  0.9792,    Adjusted R-squared:  0.9791
F-statistic: 1.288e+04 on 23 and 6304 DF, p-value: < 2.2e-16
```

- Teacher Quality

```
Call:
lm(formula = Exam_Score ~ dummyParental_I + dummyAccess_to_Resources +
  dummyExtracurricular_Activities + dummyMotivation_Level +
  dummyInternet_Access + dummyFamily_Income + dummyPeer_Influence +
  dummyLearning_Disabilities + dummyDistance_from_Home + dummyParental_Education_Level +
  v1 + v2 + v3 + v4 + v5 + v6, data = Sample3)

Residuals:
    Min       1Q   Median       3Q      Max
-1.35083 -0.31940  0.01178  0.30420  1.20329

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    40.4770097    0.0706797   572.682 < 2e-16 ***
dummyParental_ILow -2.0022596    0.0160226  -124.964 < 2e-16 ***
dummyParental_IMedium -1.0131634    0.0128577   -78.798 < 2e-16 ***
dummyAccess_to_ResourcesLow -1.9959084    0.0159612  -125.048 < 2e-16 ***
dummyAccess_to_ResourcesMedium -0.9855739    0.0127597   -77.241 < 2e-16 ***
dummyExtracurricular_ActivitiesYes  0.5155001    0.0112460   45.839 < 2e-16 ***
dummyMotivation_LevelLow -1.0073179    0.0160057  -62.935 < 2e-16 ***
dummyMotivation_LevelMedium -0.5124901    0.0145679  -35.179 < 2e-16 ***
dummyInternet_AccessYes  1.0002530    0.0208316   48.016 < 2e-16 ***
dummyFamily_IncomeLow -0.9831283    0.0152806  -64.338 < 2e-16 ***
dummyFamily_IncomeMedium -0.4793248    0.0153053  -31.318 < 2e-16 ***
dummyPeer_InfluenceNeutral  0.5008191    0.0149583   33.481 < 2e-16 ***
dummyPeer_InfluencePositive  0.9981237    0.0148895   67.036 < 2e-16 ***
dummyLearning_DisabilitiesYes -1.0098005    0.0180649  -55.899 < 2e-16 ***
dummyDistance_from_HomeModerate  0.4966808    0.0201536   24.645 < 2e-16 ***
dummyDistance_from_HomeNear  0.9971872    0.0189028   52.753 < 2e-16 ***
dummyParental_Education_LevelHigh School -0.5000241    0.0127189  -39.314 < 2e-16 ***
dummyParental_Education_LevelPostgraduate  0.5202109    0.0158586   32.803 < 2e-16 ***
v1              0.2980735    0.0009219  323.312 < 2e-16 ***
v2              0.1998969    0.0004779  418.285 < 2e-16 ***
v3              0.0192574    0.0037579    5.125 3.07e-07 ***
v4              0.0494321    0.0003839  128.758 < 2e-16 ***
v5              0.5010355    0.0044660  112.188 < 2e-16 ***
v6              0.2315233    0.0053766   43.062 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4381 on 6304 degrees of freedom
Multiple R-squared:  0.983,    Adjusted R-squared:  0.9829
F-statistic: 1.586e+04 on 23 and 6304 DF, p-value: < 2.2e-16
```

- Peer Influence

```
Call:
lm(formula = Exam_Score ~ dummyParental_I + dummyAccess_to_Resources +
  dummyExtracurricular_Activities + dummyMotivation_Level +
  dummyInternet_Access + dummyFamily_Income + dummyTeacher_Quality +
  dummyLearning_Disabilities + dummyDistance_from_Home + dummyParental_Education_Level +
  v1 + v2 + v3 + v4 + v5 + v6, data = Sample3)

Residuals:
    Min       1Q   Median       3Q      Max
-1.30908 -0.35497  0.02686  0.35421  1.17274

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  41.5848798  0.0793920  523.792 < 2e-16 ***
dummyParental_ILow -2.0126120  0.0180767 -111.338 < 2e-16 ***
dummyParental_IMedium -1.0180693  0.0145052  -70.187 < 2e-16 ***
dummyAccess_to_ResourcesLow -2.0065326  0.0180080 -111.425 < 2e-16 ***
dummyAccess_to_ResourcesMedium -0.9817340  0.0143960  -68.195 < 2e-16 ***
dummyExtracurricular_ActivitiesYes  0.5120123  0.0126847  40.365 < 2e-16 ***
dummyMotivation_LevelLow -1.0156745  0.0180649  -56.223 < 2e-16 ***
dummyMotivation_LevelMedium -0.4954951  0.0164351  -30.149 < 2e-16 ***
dummyInternet_AccessYes  0.9809940  0.0235010  41.743 < 2e-16 ***
dummyFamily_IncomeLow -1.0077933  0.0172390  -58.460 < 2e-16 ***
dummyFamily_IncomeMedium -0.5004591  0.0172679  -28.982 < 2e-16 ***
dummyTeacher_QualityLow -0.9950747  0.0226274  -43.977 < 2e-16 ***
dummyTeacher_QualityMedium -0.5032634  0.0139315  -36.124 < 2e-16 ***
dummyLearning_DisabilitiesYes -1.0279567  0.0203794  -50.441 < 2e-16 ***
dummyDistance_from_HomeModerate  0.4998646  0.0227417  21.980 < 2e-16 ***
dummyDistance_from_HomeNear  0.9965094  0.0213292  46.720 < 2e-16 ***
dummyParental_Education_LevelHigh School -0.4868529  0.0143438  -33.942 < 2e-16 ***
dummyParental_Education_LevelPostgraduate  0.5431145  0.0178893  30.360 < 2e-16 ***
v1  0.2987622  0.0010401  287.243 < 2e-16 ***
v2  0.1997332  0.0005390  370.595 < 2e-16 ***
v3  0.0123507  0.0042390   2.914 0.00359 ***
v4  0.0489237  0.0004332  112.929 < 2e-16 ***
v5  0.4996568  0.0050387  99.164 < 2e-16 ***
v6  0.2334829  0.0060663  38.489 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4943 on 6304 degrees of freedom
Multiple R-squared:  0.9784,    Adjusted R-squared:  0.9783
F-statistic: 1.24e+04 on 23 and 6304 DF,  p-value: < 2.2e-16
```

- Learning Disabilities

```
Call:
lm(formula = Exam_Score ~ dummyParental_I + dummyAccess_to_Resources +
  dummyExtracurricular_Activities + dummyMotivation_Level +
  dummyInternet_Access + dummyFamily_Income + dummyTeacher_Quality +
  dummyPeer_Influence + dummyDistance_from_Home + dummyParental_Education_Level +
  v1 + v2 + v3 + v4 + v5 + v6, data = Sample3)

Residuals:
    Min       1Q   Median       3Q      Max
-1.60703 -0.23178  0.04292  0.32218  0.91050

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  40.7440585  0.0718180  567.324 < 2e-16 ***
dummyParental_ILow -1.9815505  0.0161544 -122.663 < 2e-16 ***
dummyParental_IMedium -0.9975335  0.0129606  -76.967 < 2e-16 ***
dummyAccess_to_ResourcesLow -2.0084796  0.0160902 -124.826 < 2e-16 ***
dummyAccess_to_ResourcesMedium -0.9925799  0.0128629  -77.166 < 2e-16 ***
dummyExtracurricular_ActivitiesYes  0.5115377  0.0113379  45.118 < 2e-16 ***
dummyMotivation_LevelLow -1.0203024  0.0161398  -63.216 < 2e-16 ***
dummyMotivation_LevelMedium -0.4889606  0.0146860  -33.294 < 2e-16 ***
dummyInternet_AccessYes  0.9930419  0.0209983  47.291 < 2e-16 ***
dummyFamily_IncomeLow -0.9769693  0.0154026  -63.429 < 2e-16 ***
dummyFamily_IncomeMedium -0.4822564  0.0154291  -31.256 < 2e-16 ***
dummyTeacher_QualityLow -1.0098510  0.0202179  -49.948 < 2e-16 ***
dummyTeacher_QualityMedium -0.5043493  0.0124488  -40.514 < 2e-16 ***
dummyPeer_InfluenceNeutral  0.5065012  0.0150815  33.584 < 2e-16 ***
dummyPeer_InfluencePositive  1.0159277  0.0150089  67.688 < 2e-16 ***
dummyDistance_from_HomeModerate  0.5082333  0.0203183  25.014 < 2e-16 ***
dummyDistance_from_HomeNear  1.0083192  0.0190572  52.910 < 2e-16 ***
dummyParental_Education_LevelHigh School -0.5076536  0.0128199  -39.599 < 2e-16 ***
dummyParental_Education_LevelPostgraduate  0.5183257  0.0159880  32.420 < 2e-16 ***
v1  0.2989718  0.0009293  321.729 < 2e-16 ***
v2  0.2003967  0.0004817  416.044 < 2e-16 ***
v3  0.0143304  0.0037879   3.783 0.000156 ***
v4  0.0493094  0.0003872  127.349 < 2e-16 ***
v5  0.4990057  0.0045017  110.848 < 2e-16 ***
v6  0.2316084  0.0054201  42.731 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4416 on 6303 degrees of freedom
Multiple R-squared:  0.9827,    Adjusted R-squared:  0.9827
F-statistic: 1.495e+04 on 24 and 6303 DF,  p-value: < 2.2e-16
```


- Distance from Home

```
Call:
lm(formula = Exam_Score ~ dummyParental_I + dummyAccess_to_Resources +
    dummyExtracurricular_Activities + dummyMotivation_Level +
    dummyInternet_Access + dummyFamily_Income + dummyTeacher_Quality +
    dummyPeer_Influence + dummyLearning_Disabilities + dummyParental_Education_Level +
    v1 + v2 + v3 + v4 + v5 + v6, data = Sample3)

Residuals:
    Min       1Q   Median       3Q      Max
-1.50045 -0.30067  0.02169  0.34706  1.03199

Coefficients:
(Intercept)                41.5653768    0.0733009    567.052    < 2e-16 ***
dummyParental_ILow        -1.9954988    0.0169099   -118.008    < 2e-16 ***
dummyParental_IMedium     -1.0065410    0.0135675    -74.188    < 2e-16 ***
dummyAccess_to_ResourcesLow -2.0086544    0.0168425   -119.261    < 2e-16 ***
dummyAccess_to_ResourcesMedium -0.9739015    0.0134622    -72.343    < 2e-16 ***
dummyExtracurricular_ActivitiesYes 0.4969984    0.0118643    41.890    < 2e-16 ***
dummyMotivation_LevelLow  -1.0117973    0.0168933   -59.893    < 2e-16 ***
dummyMotivation_LevelMedium -0.4919323    0.0153738   -31.998    < 2e-16 ***
dummyInternet_AccessYes    0.9910506    0.0219808    45.087    < 2e-16 ***
dummyFamily_IncomeLow     -1.0010200    0.0161234   -62.085    < 2e-16 ***
dummyFamily_IncomeMedium  -0.4925516    0.0161472   -30.504    < 2e-16 ***
dummyTeacher_QualityLow   -0.9950187    0.0211618   -47.020    < 2e-16 ***
dummyTeacher_QualityMedium -0.5031765    0.0130305   -38.615    < 2e-16 ***
dummyPeer_InfluenceNeutral  0.5055854    0.0157870    32.025    < 2e-16 ***
dummyPeer_InfluencePositive 0.9970466    0.0157121    63.457    < 2e-16 ***
dummyLearning_DisabilitiesYes -1.0078909    0.0190611   -52.877    < 2e-16 ***
dummyParental_Education_LevelHigh School 0.4897836    0.0134199   -36.497    < 2e-16 ***
dummyParental_Education_LevelPostgraduate 0.5216646    0.0167362    31.170    < 2e-16 ***
v1                        0.2974107    0.0009727    305.756    < 2e-16 ***
v2                        0.2005176    0.0005042    397.708    < 2e-16 ***
v3                        0.0173489    0.0039654     4.375    1.23e-05 ***
v4                        0.0496905    0.0004053    122.606    < 2e-16 ***
v5                        0.5053737    0.0047118    107.298    < 2e-16 ***
v6                        0.2379802    0.0056739    41.943    < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4623 on 6304 degrees of freedom
Multiple R-squared:  0.9811,    Adjusted R-squared:  0.981
F-statistic: 1.421e+04 on 23 and 6304 DF,  p-value: < 2.2e-16
```

- Parental Education Level

```
Call:
lm(formula = Exam_Score ~ dummyParental_I + dummyAccess_to_Resources +
    dummyExtracurricular_Activities + dummyMotivation_Level +
    dummyInternet_Access + dummyFamily_Income + dummyTeacher_Quality +
    dummyPeer_Influence + dummyLearning_Disabilities + dummyDistance_from_Home +
    v1 + v2 + v3 + v4 + v5 + v6, data = Sample3)

Residuals:
    Min       1Q   Median       3Q      Max
-1.12649 -0.38121 -0.04034  0.35130  1.44975

Coefficients:
(Intercept)                40.6960806    0.0821135    495.608    < 2e-16 ***
dummyParental_ILow        -1.9801352    0.0185539   -106.723    < 2e-16 ***
dummyParental_IMedium     -1.0100314    0.0148849    -67.856    < 2e-16 ***
dummyAccess_to_ResourcesLow -2.0029934    0.0184804   -108.385    < 2e-16 ***
dummyAccess_to_ResourcesMedium -0.9730566    0.0147708    -65.877    < 2e-16 ***
dummyExtracurricular_ActivitiesYes 0.5074904    0.0130211    38.975    < 2e-16 ***
dummyMotivation_LevelLow  -1.0058136    0.0185353   -54.265    < 2e-16 ***
dummyMotivation_LevelMedium -0.4899251    0.0168637   -29.052    < 2e-16 ***
dummyInternet_AccessYes    0.9961821    0.0241112    41.316    < 2e-16 ***
dummyFamily_IncomeLow     -0.9884645    0.0176925   -55.869    < 2e-16 ***
dummyFamily_IncomeMedium  -0.4799784    0.0177207   -27.086    < 2e-16 ***
dummyTeacher_QualityLow   -1.0047357    0.0232204   -43.269    < 2e-16 ***
dummyTeacher_QualityMedium -0.5123971    0.0142972   -35.839    < 2e-16 ***
dummyPeer_InfluenceNeutral 0.4928542    0.0173139    28.466    < 2e-16 ***
dummyPeer_InfluencePositive 1.0077757    0.0172353    58.472    < 2e-16 ***
dummyLearning_DisabilitiesYes -1.0231896    0.0209129   -48.926    < 2e-16 ***
dummyDistance_from_HomeModerate 0.5013755    0.0233371    21.484    < 2e-16 ***
dummyDistance_from_HomeNear 0.9964926    0.0218875    45.528    < 2e-16 ***
v1                        0.2977325    0.0010673    278.955    < 2e-16 ***
v2                        0.2008858    0.0005531    363.201    < 2e-16 ***
v3                        0.0200779    0.0043506     4.615    4.01e-06 ***
v4                        0.0491631    0.0004446    110.569    < 2e-16 ***
v5                        0.5028117    0.0051705     97.246    < 2e-16 ***
v6                        0.2229593    0.0062225    35.831    < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5072 on 6304 degrees of freedom
Multiple R-squared:  0.9772,    Adjusted R-squared:  0.9771
F-statistic: 1.176e+04 on 23 and 6304 DF,  p-value: < 2.2e-16
```

In each hypothesis test, adjusted R^2 decreased, and SSE increased.

Therefore, the data supports that each coefficient is significant to our model, so H_0 is rejected for all tests.

ANOVA TABLE

Analysis of Variance Table

Response: Exam_Score

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
dummyParental_I	2	2372	1186	1.1886e+04	<2e-16	***
dummyAccess_to_Resources	2	2744	1372	1.3754e+04	<2e-16	***
dummyExtracurricular_Activities	1	377	377	3.7755e+03	<2e-16	***
dummyMotivation_Level	2	758	379	3.7975e+03	<2e-16	***
dummyInternet_Access	1	305	305	3.0543e+03	<2e-16	***
dummyFamily_Income	2	709	355	3.5537e+03	<2e-16	***
dummyTeacher_Quality	2	502	251	2.5140e+03	<2e-16	***
dummyPeer_Influence	2	852	426	4.2695e+03	<2e-16	***
dummyLearning_Disabilities	1	813	813	8.1472e+03	<2e-16	***
dummyDistance_from_Home	2	844	422	4.2308e+03	<2e-16	***
dummyParental_Education_Level	2	1182	591	5.9229e+03	<2e-16	***
v1	1	19991	19991	2.0039e+05	<2e-16	***
v2	1	33230	33230	3.3309e+05	<2e-16	***
v3	1	0	0	7.9300e-02	0.7783	
v4	1	3092	3092	3.0998e+04	<2e-16	***
v5	1	2446	2446	2.4518e+04	<2e-16	***
v6	1	369	369	3.7019e+03	<2e-16	***
Residuals	6302	629	0			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

All variables are significant to the model since their p-values in the ANOVA table are all below 0.05.

Confidence Intervals

	2.5 %	97.5 %
(Intercept)	40.76583609	40.96732030
dummyParental_ILow	-2.01056502	-1.96526443
dummyParental_IMedium	-1.02504527	-0.98869862
dummyAccess_to_ResourcesLow	-2.02719267	-1.98207269
dummyAccess_to_ResourcesMedium	-1.00645879	-0.97038863
dummyExtracurricular_ActivitiesYes	0.48935293	0.52114768
dummyMotivation_LevelLow	-1.03799627	-0.99273688
dummyMotivation_LevelMedium	-0.51910114	-0.47791654
dummyInternet_AccessYes	0.96977361	1.02865713
dummyFamily_IncomeLow	-1.01135889	-0.96816268
dummyFamily_IncomeMedium	-0.51236185	-0.46909401
dummyTeacher_QualityLow	-1.03892350	-0.98222941
dummyTeacher_QualityMedium	-0.52212193	-0.48721345
dummyPeer_InfluenceNeutral	0.48042166	0.52271343
dummyPeer_InfluencePositive	0.98361190	1.02570311
dummyLearning_DisabilitiesYes	-1.03600619	-0.98494331
dummyDistance_from_HomeModerate	0.47263475	0.52961152
dummyDistance_from_HomeNear	0.97960960	1.03304912
dummyParental_Education_LevelHigh school	-0.51620901	-0.48025692
dummyParental_Education_LevelPostgraduate	0.49938263	0.54421578
v1	0.29701028	0.29961630
v2	0.19928281	0.20063368
v3	0.01176550	0.02238831
v4	0.04894808	0.05003389
v5	0.49472126	0.50734514
v6	0.22828848	0.24348897

The 95% confidence intervals for all coefficients are very narrow, and none of them crosses zero, indicating highly precise estimates. This confirms that every predictor in the model has a statistically significant effect on exam scores.

Conclusion

The analysis identified Attendance (v2), Hours Studied (v1), and Motivation Level as the most important predictors of student exam scores, with strong positive effects. Key findings confirm that academic performance is not driven by a single factor but is a complex outcome shaped by a combination of behavioural factors (study habits and attendance), socioeconomic and resource-related factors (family income, parental involvement, internet access, and teacher quality), and psycho-social factors (motivation and peer influence). The final stepwise regression model successfully integrated these predictors, explaining over 98% of the variance in exam scores, with all variables being statistically significant.

However, the study has several limitations: the model's exceptionally high R^2 may suggest overfitting, despite cross-validation not being fully detailed; the use of self-reported data could introduce measurement bias; and the analysis is correlational, preventing causal inferences. Additionally, variables such as sleep hours showed negligible impact, which may reflect data limitations rather than true irrelevance.

For future research, it is recommended to:

- Validate the model on an independent dataset to assess generalizability and reduce overfitting risk.
- Incorporate interaction terms or nonlinear relationships to explore more complex dynamics among predictors.
- Collect longitudinal data to better establish causality and track performance trends over time; and
- Include additional contextual factors such as school funding, classroom size, or mental health indicators to provide a more holistic understanding of academic success.